

Acknowledgements

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1 Introduction

A compound that enters a cell changes what that cell transcribes. Which genes move, by how much, and in which direction is the most direct readout available of what the compound actually does — not what it was designed to bind, but what the cell does in response. Reading that response is routine. Reading it for every combination that might matter is not. A compound behaves differently at different doses, and differently again depending on the genetic background of the cell receiving it, so the space to be explored is the product of three large sets rather than any one of them. Even the largest screens sample a small corner of it, and every combination left untested is a question left open.

This is the gap prediction is meant to close. If the transcriptional response to a treatment could be computed rather than measured, the untested corners of that space would become inspectable, and the experiments actually run could be chosen because they are informative rather than because they happened to fit on the plate. The task, stated plainly, is this: given the transcriptome of an untreated cell and the identity of a drug, predict the transcriptome that cell would have shown had the drug been applied.

The task is harder than it sounds, for a reason that is easy to state and impossible to engineer around. The ground truth is a counterfactual. Sequencing destroys the cell it reads, so the treated and untreated profiles of the same cell can never both be observed, and a prediction can only ever be checked against a different cell that received the drug. Everything downstream inherits this. It is why paired training data must be assembled by matching cells rather than by tracking them, why evaluation is usually framed at the level of populations rather than individuals, and why distinguishing a real drug effect from ordinary cell-to-cell variation requires care rather than optimism.

Assembling data at the scale this problem demands became possible only recently. Two decades of falling sequencing costs, rising throughput and increasingly clever sample multiplexing turned single-cell profiling from a delicate experiment into something that can be run industrially, and Tahoe-100M [1] is what the end of that trajectory looks like: roughly a hundred million single-cell transcriptomes spanning some 1,100 small molecules across 50 cancer cell lines. Every compound and every cell line appears not as a summary statistic but as a cloud of individual cells.

That resolution changes which questions can be asked. The obvious one is conditional — what does this compound do to this cellular context, rather than what does it do on average across contexts that may have little in common. The less obvious question is whether the first has an answer. Because many conditions were run more than once, it becomes possible to measure how much of an apparent drug effect reproduces across replicates and how much is simply the variability of the assay. That second measurement turns out to matter as much as the first, and a good deal of what follows rests on the observation that standard evaluations quietly confuse the two.

The other development behind this thesis has nothing to do with cells. Over the same period, pretrained transformer language models became unreasonably good and unreasonably general, and the engineering poured into them — architectures, training recipes, scaling laws, the entire surrounding toolchain — came to exceed what any single scientific field could plausibly assemble for itself. That created a standing temptation: if a problem can be made to look like text, it inherits all of that work for free.

Cell2Sentence [2] takes the temptation about as literally as it can be taken. Rank the genes in a cell from most to least expressed, write their names out in that order, and the cell becomes a sentence — ordinary text that an ordinary language model can be fine-tuned on. The encoding is unsentimental about biology, discarding how much each gene is expressed and keeping only the ordering, on the argument that rank retains enough of the signal for the rest to be reconstructed. In exchange it requires no new architecture and no new training machinery, only cells written in a format that existing models already consume.

C2S-Scale [3] develops that idea into a family of models. Larger backbones are pre-trained on over 50 million cells drawn from 825 single-cell datasets in CELLxGENE and the Human Cell Atlas, and then fine-tuned across a spread of tasks: predicting cell type and tissue, generating cells conditioned on a label, captioning clusters, answering questions about a dataset in natural language, and predicting perturbation responses. That last task is the one this thesis builds on, and it is worth being specific about its scope. Their perturbation work covers cytokine stimulation in peripheral blood cells, and, for small molecules, bulk L1000 profiles and the SciPlex3 panel. The single-cell results are scored as distributions — how closely the generated population for a condition resembles the observed one — rather than condition by condition. The checkpoint used throughout this thesis, **C2S-Scale-Pythia-1b-pt**, is the pretrained model from that family: atlas data only, with no drug-treated cells among it.

What we do is extend that line of work to chemical perturbation at the scale Tahoe-100M now makes available, and evaluate it as a prediction problem in the strict sense — condition by condition, against controls chosen so that a model which ignores the drug cannot pass. The question is whether a cell-sentence model fine-tuned on this atlas learns what a particular compound does to a particular cell.

Two things are worth keeping straight about what the model has actually seen, since the scale of the atlas invites an easy overstatement. Its pretraining is large but drug-free: tens of millions of cells, none of them treated with anything. And the fine-tuning does not consume Tahoe entire — it draws a sample of a few hundred thousand treated cells from it, which is ample for the task but is not a hundred million. Whatever the model learns about drugs, it learns during fine-tuning, from that sample.

It does learn something, and the something is the difficulty. The model acquires a good account of what a treated cell looks like *in general* — the shared shape that

perturbed transcriptomes have in common, which is substantial and largely the same whichever compound produced it. What it does not acquire is the part that distinguishes one compound from another. Change the drug named in the prompt while holding the input cell fixed, and the prediction barely moves; on the calibrated metric the model is statistically indistinguishable from a baseline that copies the untreated cell and is told nothing about any drug at all.

A second comparison is worth separating from that one, because it is weaker but more informative. A per-drug average, estimated from the same compound in *other* cell lines, also outperforms the model — but this is not a naive baseline. It is very nearly the best any method can do at the part of the problem it addresses, scoring 0.963 against a replicate ceiling of 0.968, and being close to it would be a real achievement rather than a low bar. What it cannot do is condition on the cell in front of it, which is precisely why the comparison is useful: it separates the drug’s average effect from the part that depends on context. Section 4.5 measures that split, and finds the second part worth roughly half the drug-specific signal — reached by the average, by the model, and by the literature alike, which is to say not at all.

A negative result earns its place only if it can be made specific, and that is the organising principle of what follows. We establish first that the failure is real rather than an artifact of how this field measures success — a distinction less pedantic than it sounds, since the standard metric can be saturated by a predictor that knows nothing about drugs. We then locate where in the pipeline the failure originates, show that the location implies a repair, apply it, and finally bound how much of the problem any method of this kind could recover even if that repair were perfect. The thesis therefore ends not in closure but in a number: a measured fraction of the drug-specific signal claimed by nothing in the literature, by no baseline we could construct, and by none of our own models.

Four questions carry that argument, in order:

1. **Is the evaluation trustworthy?** Before asking whether a model is any good, one has to ask whether the metric could tell. Can a predictor carrying no drug information score well on the measures the field has standardised on, and if so, which measures survive?
2. **Does the model use the drug at all?** Holding the input cell fixed and changing only the compound it is told about, does the prediction move — and does it move in the direction the real response moved?
3. **If not, where does the failure live?** Is the drug missing from the data, missing from the model’s internal representation, or present in that representation but ignored by the process that writes the output? Each answer implies a different repair, and only one of them is cheap.

4. **How much is recoverable at all?** A per-drug average is a strong baseline. How much of the drug-specific signal does it capture, how much is genuinely conditional on cellular context, and how much does any method — ours or anyone else’s — actually reach?

The contributions follow the same order. We audit the calibration of the standard evaluation and show that it can be saturated by a predictor carrying no drug information whatsoever, which bears on any work built from the same family of metrics. We establish drug-blindness on the one metric that survives that audit, using several independent instruments so that no single measurement carries the claim alone. We identify the target encoding — not the data, the architecture, or the training objective — as the binding constraint, and derive from that diagnosis a re-encoding of the prediction target which produces the first measurable drug effect in this project and generalises to drug-and-cell-line pairings never seen during training. And we separate the per-drug main effect from the drug-by-cell-line interaction, showing that the interaction accounts for roughly half the drug-specific signal and is reached by neither the lookup baseline nor the model.

The thesis proceeds as follows. Section 2 places the work against single-cell foundation models, the perturbation-prediction literature, and an unsettled dispute within it about whether deep models beat trivial baselines at all — a dispute this thesis adjudicates in one corner, using the instruments of the side it ultimately argues against. Section 3 covers the data, the model, the calibration framework, and, at some length because they carry the argument, the controls that make each negative result interpretable. Section 4 reports the findings in the order the argument needs rather than the order in which they arrived. Section 5 sets out the scope honestly, separating what is closed by measurement from what is merely untested, and Section 6 concludes.

2 Literature review

2.1 From text foundation models to single-cell foundation models

The models this thesis builds on descend from a template established in natural language processing. A transformer is pretrained on a very large unlabelled text corpus with a single objective — predict the next token — and the resulting network is then adapted to specific tasks by fine-tuning or prompting. The GPT line of models made the case that this recipe scales: capability on tasks never explicitly trained for improves steadily with model size, data and compute, so a single pretrained backbone can serve as a general substrate rather than a task-specific tool. What made the recipe portable is that it assumes almost nothing about language. It requires only that the data be expressible as sequences of discrete tokens, and that some tokens be predictable from others.

That portability is what invited the transfer into biology. If a cell can be written as a sequence, the same machinery applies unchanged. scGPT [4] and Geneformer [5] both take this route, treating a cell’s expression profile as a sequence of gene tokens and pretraining on tens of millions of cells before transferring to classification, integration and perturbation tasks. The ambition is usually stated as some version of the *virtual cell* [6]: a model whose simulated responses to intervention are trustworthy enough to substitute for an experiment.

The interesting variation among these models is not architectural but representational. A cell is a vector of expression values over some twenty thousand genes; a transformer consumes a sequence of discrete tokens. Something must be discretised or discarded in between, and what gets discarded bounds what the model can learn. The approach this thesis uses makes that choice unusually explicit.

2.2 Cell sentences, and the model this thesis fine-tunes

Cell2Sentence [2] takes the most literal route available: rank a cell’s genes by expression and write their symbols out in descending order. The cell becomes a *cell sentence*, which is text in the ordinary sense — a string that a tokeniser will happily consume:

MALAT1 ACTB TMSB4X B2M EEF1A1 RPL13A FTL TPT1 RPS27 GAPDH ...

Nothing marks this as biological. The information is carried entirely by which symbols appear and in what order, and a language model fine-tuned on such strings is doing exactly what it does on prose: predicting the next token given the previous ones.

The encoding keeps rank and discards magnitude. C2S defends the loss with a reconstruction argument: expression can be recovered from rank through a fitted log-linear relationship $e_i = a_d \log(r_i) + b_d$, which achieves $R^2 = 0.815$ against the original values

on peripheral blood data. On that basis they argue rank encodes expression reversibly enough across datasets for the discarded magnitude not to matter for downstream use. It is worth noting what kind of criterion this is — reconstruction fidelity is a statement about *absolute* accuracy, how close the recovered profile is to the true one, which is not automatically the same as preserving whatever distinguishes one experimental condition from another.

Understanding what the model does with such a sentence requires a little of the underlying architecture, because several of its properties matter later. C2S-Scale builds on decoder-only transformers of the GPT family, and the checkpoint used here is built on Pythia [7], a suite of such models released with open weights and training data. A decoder-only transformer factorises the probability of a sequence into a product of next-token predictions,

$$p(x_1, \dots, x_T) = \prod_{t=1}^T p(x_t \mid x_{<t}), \quad (1)$$

and is trained by minimising the cross-entropy between each predicted distribution and the token that actually occurred. Each token is first mapped to a vector by an embedding matrix, and those vectors form what is usually called the *residual stream*: a running representation, one vector per position, which every layer reads from and writes back into additively. A layer comprises multi-head self-attention, which lets each position gather information from earlier positions and is *causally masked* so that position t attends only to positions $\leq t$, followed by a position-wise feed-forward network. Both are wrapped in normalisation and added back to the stream rather than replacing it. After the final layer an unembedding matrix projects the stream to one score per vocabulary entry, and a softmax turns those scores into the next-token distribution.

Three properties of that design are worth noting now, since later sections depend on them. The residual stream is additive and shared, so a representation can be read at a chosen depth, modified, and the remaining computation allowed to proceed on the modified value — which is what makes interpretability interventions of the kind used in Section 4.3 possible at all. The training loss is a sum of per-token cross-entropies that weights every position equally, so a target whose tokens are largely shared between conditions gives any individual condition only a small share of the gradient. And conditioning is mediated entirely by attention across a causal mask, so information supplied early in a prompt must be reached from every later position at which it is needed.

C2S-Scale [3] develops Cell2Sentence into a family of models on this architecture, spanning backbones from Pythia at 410 million and 1 billion parameters up to Gemma-2 at 27 billion. Pretraining uses over 50 million human and mouse cells drawn from 825 single-cell datasets in CELLxGENE and the Human Cell Atlas. The pretrained models are then fine-tuned across a range of tasks: predicting cell type and tissue, generating cells conditioned on a label, captioning clusters, answering natural-language questions about

a dataset, and predicting perturbation responses.

The perturbation task is the one this thesis extends, and its published scope is worth stating precisely. C2S-Scale evaluates it on three benchmarks: cytokine stimulation of peripheral blood cells, bulk L1000 profiles, and the SciPlex3 small-molecule panel. In each the model receives an untreated profile together with a description of the treatment and produces the treated profile, so the setup is conditional rather than label-only generation. Two features of it are relevant to the comparison this thesis draws. Pairing is by group rather than by cell — for the bulk benchmark, treated and untreated samples are matched by cell line name — which is unavoidable given that sequencing destroys the cell it reads. And the single-cell results are scored as distributions: for each condition the model generates as many cells as were observed, and the generated population is compared against the observed one, which asks whether the model produces a plausible population for a condition rather than whether it distinguishes that condition from another.

The checkpoint fine-tuned in every experiment below is **C2S-Scale-Pythia-1b-pt**, the pretrained member of that family, whose training corpus is atlas data alone and contains no drug-treated cells.

2.3 The dispute: do these models beat uninformative baselines?

The field contains an unresolved and unusually sharp disagreement about whether any of this works, and this thesis is best read as an attempt to settle one corner of it.

On one side sits a series of benchmarks reporting that deep perturbation models do not beat trivial predictors. Ahlmann-Eltze, Huber and Anders [8] find that deep-learning prediction of genetic perturbation effects does not outperform simple linear baselines. Csendes et al. [9] reach the same conclusion for foundation cell models on post-perturbation RNA-seq, and Bendidi et al. [10] put it most bluntly in a title, reporting that a single PCA still rules them all. Kernfeld et al. [11] concur in a broad comparison of expression-forecasting methods. The pattern is not confined to transcriptomic readouts: on scalar drug sensitivity, Hauptmann and Kramer [12] hold the experimental protocol fixed across architectures and find that the measured differences between them largely dissolve, and DrEval [13] finds that deep models barely outperform a naive predictor built from mean drug and cell-line effects. The recurring result is that a per-perturbation mean is remarkably hard to beat.

Against this, Miller et al. [14] argue that these conclusions “largely stem from limitations of benchmarking metrics, not from the models themselves”, and the argument is methodological rather than rhetorical. They formalise the use of positive and negative controls in perturbation benchmarking and introduce a new positive control, the *interpolated duplicate*, which blends a technical duplicate with the mean baseline using weights drawn from per-gene differential-expression significance — giving a reference point of known and

tunable quality. They then define a quantitative measure of whether a metric can see that quality at all, the *Dynamic Range Fraction* (DRF), and apply it across 14 Perturb-seq datasets and 13 metrics. Widely used measures such as MSE and control-referenced Pearson correlation turn out to be poorly calibrated, while weighted and rank-based alternatives — weighted MSE, weighted $R^2\Delta$ [15], and Normalised Inverse Rank [16] — are consistently calibrated. Evaluated on the metrics that survive, they report, deep models *do* beat mean, control and linear baselines.

What is actually in dispute is narrower than it first appears, because both sides agree the standard metrics are defective. The disagreement is about what a corrected evaluation would reveal. One reading is that the models were always capturing real biology and the broken metrics were hiding it, so better measurement should expose genuine skill. The other is that the models were never capturing much, and better measurement will simply show that failure more clearly and more honestly. Put that way the question is empirical rather than interpretive — but it is settled only within the setting where it is tested, and the setting of this thesis has been tested by neither camp. Miller et al. and Ahlmann-Eltze et al. both study **genetic** perturbation, where an intervention is a designed, near-binary knockout with a named target gene. DrEval studies **drug sensitivity** as a scalar potency such as IC_{50} , predicted from cell-line omics. None of them studies a full transcriptomic response to a small molecule — which acts at a dose and typically engages several proteins at once — at single-cell resolution, produced by an autoregressive generative model.

The position taken here is to adopt the proponents’ instrument rather than argue with it. Section 4.1 ports the DRF framework onto drug perturbation and uses it to decide which metrics are admissible in this setting before any model is scored, so that the evaluation is chosen by the sceptics’ opponents rather than by us. Whatever the models then turn out to do, the measurement cannot be dismissed by the objection Miller et al. themselves raise.

2.4 Six ways a drug-response benchmark can mislead

DrEval [13] deserves separate treatment, because beyond its verdict it supplies a diagnostic vocabulary that this thesis adopts throughout. Auditing drug response prediction, it identifies six obstacles to clinical translation: non-reproducible models, data leakage producing weak generalisation, pseudoreplication, biased evaluation metrics, missing ablation studies, and inconsistent response data. Four of them shape the methodology of Section 3 directly.

Pseudoreplication, in Hurlbert’s original sense [17], is treating repeated measurements of one unit as independent observations. DrEval observes that GDSC2’s 395,025 measurements descend from only 969 cell lines and 287 drugs, so an interval computed over measurements is badly overconfident about how much independent evidence exists. Our

data has the same shape — hundreds of thousands of cells resolving to a few thousand conditions across 50 cell lines — and every confidence interval in this thesis is consequently a clustered bootstrap over cell lines rather than over cells.

Biased evaluation they frame as a version of Simpson’s paradox. Because different drugs act at vastly different concentrations, most of the variance in a response dataset lies *between* drugs, so a model that learns nothing but each drug’s mean will explain most of it. Their illustration is stark: a global Pearson correlation of 0.90 alongside an average within-drug Pearson of 0.01. Section 3.3 explains why a discrimination metric evaluated *within* a comparison set is immune to this by construction.

Their treatment of splits is the part we borrow most directly. DrEval names four, each corresponding to a different application: Leave-Pairs-Out for missing-value imputation, Leave-Cell-Lines-Out for personalised medicine, Leave-Drugs-Out for drug design, and Leave-Tissue-Out for repurposing. Our holdout maps onto two of these — `unseen_combo` is Leave-Pairs-Out and `unseen_drug` is Leave-Drugs-Out — and Section 3.8 uses their names. The mapping carries a warning we take seriously. Citing Partin et al. [18], DrEval reports that good Leave-Pairs-Out and Leave-Cell-Lines-Out performance is reachable simply by predicting each drug’s average response across the training cell lines, so a strong result on those splits demands that the per-drug average be measured explicitly rather than assumed away.

Their remedy for missing ablations is randomisation and robustness testing. The scramble ablation of Section 3.4 is such a test, specialised to the drug: hold every byte of the prompt fixed except the compound, and see whether anything downstream notices.

2.5 Connectivity mapping and signature transfer

The Connectivity Map programme [19, 20] built an entire drug-discovery methodology on a specific premise: that a compound induces a characteristic transcriptional signature stable enough across cellular contexts that querying a reference database by signature similarity is informative. The L1000 platform [20] operationalised this at scale, and its 978-gene landmark set defines the gene panel used here, intersected with Tahoe’s measured genes to give 946.

The premise — that a drug’s signature transfers across cell lines — is normally assumed rather than measured, and how much of it holds determines whether a conditional model has anything left to predict once a per-drug constant is accounted for. Section 4.5 measures it directly, corrected for measurement noise, at single-cell resolution.

2.6 Decodable, used, and the global workspace

Linear probing establishes that information is *present* in a representation, not that it is *used* by the computation, and the distinction is central to the analysis in Section 4.3.

Elazar et al. [21] make the methodological case for keeping the two apart: probe accuracy licenses no conclusion about behaviour, and the way to ask whether a property is used is to remove it from the representation and measure what the model then does. Their amnesic-probing design is the one adopted here, with the addition of a matched control, since an intervention that fails to move the output proves nothing unless a comparable intervention does.

A recent line of work gives that distinction a sharper form. Gurnee et al. [22] report that language models maintain a small privileged set of representations — which they term the *J-space*, identified by a Jacobian Lens that measures the average linearised effect of an activation on the model’s likelihood of producing a particular token — and argue it functions as a global workspace analogous to conscious access. The J-space is strikingly small: roughly ten to twenty-five concepts are active at any moment, typically accounting for under 10% of activation variance. Its defining property is causal rather than correlational, and they establish it by intervention: swapping one active workspace vector for another changes the model’s answer to match. Content in the J-space drives behaviour; content outside it may be richly represented and still inert with respect to what the model actually emits.

That framing matters here because it supplies both a hypothesis and an experimental design. The hypothesis is that a property can be decodable from the residual stream at high accuracy while sitting outside the workspace the generation readout consults, in which case no amount of encoding it more explicitly will change the output. The design is the swap itself: overwrite one candidate representation with another and ask whether the model’s behaviour follows. Section 4.3 constructs a domain analogue of exactly this test, substituting a drug’s learned representation for another drug’s and measuring whether the emitted profile moves toward the substituted drug.

The related account of why a model might allocate its representations this way comes from Elhage et al. [23], who show that networks reserve dedicated directions for features the computation needs on every forward pass and push the remainder into superposition, where a feature remains linearly recoverable without occupying an axis of its own. Read together, the two suggest an asymmetry worth testing: a property required to emit any plausible output should sit in the causally active set, while one needed only for occasional, context-dependent composition need not.

2.7 Optimal transport for state matching

Predicting a treated cell from an untreated one runs into an obstacle that is not statistical but physical: sequencing destroys the cell it measures. No cell is ever observed both before and after treatment, so there is no ground-truth correspondence between the control population and the treated population — only two clouds of points in expression space,

sampled from the same well at different times. Any construction of paired training data must therefore *infer* a correspondence, and optimal transport is the standard framework for inferring one.

The problem in its classical form asks for the cheapest way to move one distribution of mass onto another. Monge’s original formulation seeks a map $T : \mathcal{X} \rightarrow \mathcal{Y}$ pushing a source measure μ onto a target ν , so that $T_{\#}\mu = \nu$, while minimising

$$\int_{\mathcal{X}} c(x, T(x)) \, d\mu(x), \quad (2)$$

where c is a ground cost, typically squared Euclidean distance. This is intuitive — every source point is assigned one destination — but it is frequently ill-posed, because a deterministic map cannot split mass, and no such map exists when the target is more spread out than the source. Kantorovich’s relaxation resolves this by replacing the map with a *coupling*: a joint distribution $\pi \in \Pi(\mu, \nu)$ whose marginals are μ and ν , giving

$$\min_{\pi \in \Pi(\mu, \nu)} \int_{\mathcal{X} \times \mathcal{Y}} c(x, y) \, d\pi(x, y). \quad (3)$$

A coupling may send fractions of a source point to several destinations, which is what makes the relaxed problem always solvable and appropriate for noisy, overlapping empirical distributions. When c is the p -th power of a metric the optimal value defines the p -Wasserstein distance $W_p(\mu, \nu)$; the case $p = 2$ is the one used here, and its geodesics give a principled way to interpolate between distributions.

Solved exactly, the discrete problem is a linear program with roughly $O(n^3 \log n)$ cost, which is prohibitive at single-cell scale. Entropic regularisation makes it tractable. Adding a negative entropy term to the objective,

$$\min_{\pi \in \Pi(\mu, \nu)} \langle C, \pi \rangle - \varepsilon H(\pi), \quad (4)$$

renders the problem strictly convex, so the optimum is unique and takes the scaling form $\pi = \text{diag}(u) K \text{diag}(v)$ with Gibbs kernel $K_{ij} = \exp(-c(x_i, y_j)/\varepsilon)$. The Sinkhorn–Knopp algorithm recovers u and v by alternating matrix–vector products [24], which are cheap and parallelise well on a GPU.

The regularisation parameter ε is not merely a computational convenience, and it is the reason this framework appears in this thesis at all. It controls how sharply mass is assigned. As $\varepsilon \rightarrow 0$ the coupling concentrates and approaches the unregularised solution, in which each control cell is matched to essentially one treated cell. As ε grows the plan blurs, and in the limit every control cell is coupled equally to every treated cell — the product of the marginals. Given a coupling, the *barycentric projection* maps each control

cell x_i to the π -weighted average of its partners,

$$\hat{y}(x_i) = \frac{\sum_j \pi_{ij} y_j}{\sum_j \pi_{ij}}, \quad (5)$$

and the two extremes of ε therefore produce two familiar objects: at small ε a specific observed treated cell, and at large ε the group mean. A sweep over ε is thus a continuous interpolation between two target constructions that would otherwise have to be treated as unrelated alternatives, which is how Section 4.4 uses it.

Applied to biology, this machinery has largely been used for trajectory inference. Waddington-OT [25] treats the cell population at each time point as a distribution in expression space and recovers a probabilistic ancestor–descendant link between snapshots, on the assumption that cells follow paths minimising transport cost. Subsequent work relaxes the assumption that mass is conserved — cells divide and die, so the two populations need not have equal total mass — using unbalanced formulations built on the Wasserstein–Fisher–Rao metric. A separate line of work exploits the fact that when both measures are Gaussian the transport cost has a closed form, the Bures–Wasserstein distance, which splits cleanly into a shift in means and a change in covariance structure and so distinguishes a population that moved from one that merely spread out.

Two practical points govern how couplings are computed here. The ground cost must be evaluated in a space where Euclidean distance is meaningful, and raw counts are not such a space: they are sparse, overdispersed and dominated by library size. Couplings are therefore computed in the released Tahoe scVI latent [26], a variational autoencoder embedding that models counts with a negative-binomial likelihood and yields a compact continuous representation in which distances behave. And because the cost of a full coupling grows quadratically, transport is solved within matched groups — a single cell line on a single plate — rather than across the atlas, which is both computationally necessary and biologically correct, since a control cell from one cell line is not a plausible counterfactual partner for a treated cell from another.

3 Methods

Every design choice in this chapter was forced by a specific failure of the obvious alternative, and the chapter is organised as that chain. We need a quantity to measure (data and representation), a ruler that cannot be gamed (metrics), manipulations that isolate drug use from everything that imitates it (controls), and a way to ask where inside the model the failure sits (probing). Only then do the target constructions make sense, because each is an attempted fix for a defect the earlier sections have already localised. The two most consequential choices come last: how the data are split, which decides what any number can claim, and what the model is compared against, which decides what any number means. A reader who finishes the chapter should be able to predict the shape of the results chapter from it.

3.1 Data

The corpus is Tahoe-100M [1]: drug-treated single-cell transcriptomes across cancer cell lines, with plate-matched DMSO controls. Metadata supplies cell line (Cellosaurus ID), drug, dose (`drugname_drugconc`), plate, and a fine-grained mechanism-of-action annotation (`moa-fine`). All analysis is restricted to the 946 genes in the intersection of the L1000 landmark set [20] with Tahoe’s measured genes, and expression is library-normalised to counts-per-10k and log $1p$ -transformed.

One representational pitfall is worth recording because it is silent. Tahoe’s per-cell `genes` field holds *vocabulary token identifiers*, not positional indices into the gene metadata table. Mapping them positionally scrambles gene identity quietly — the values exceed the metadata row count, so the error surfaces only as an out-of-bounds index at the tail. All expression construction here maps `gene_metadata['token_id']` $\rightarrow$ gene symbol $\rightarrow$ panel index, and the construction is validated by a PCA cell-line probe on the resulting panel, which reaches 0.864 against a chance rate of 0.02.

Unit of analysis. A *condition* is a (drug, cell line, plate, dose) tuple, and the physical treatment assignment is the *sample* — one treated well. That distinction, cosmetic here, becomes load-bearing in Section 3.8: one well holds roughly 49 cell lines dosed together, so a condition is an observation nested inside an assignment, not an assignment. Plates are numbered within cell line rather than globally, so the batch unit is the pair (cell line, plate) and never the plate label alone — a distinction that matters for every within-plate comparison below. Dose is carried as a molar concentration resolved from `drugname_drugconc`, not as a unitless float: an earlier pipeline kept only the numeric part, which collides $1\ \mu\text{M}$ with $1\ \text{nM}$, and a direct audit of all 289 treatment samples found 289 usable molar doses with zero unit collisions — every drug in this atlas states its doses in a single unit, so the defect was a demonstrated possibility that did not materialise.

Combination samples are dropped rather than analysed as their first component.

Scale. Table 1 states the quantities every n in this thesis should be located against. Two populations are used and they are not interchangeable. The broad characterisation of the task (Section 4.2) streams a wide sample of the atlas; the residual arm (Sections 4.6 onward) works from a smaller cached subset built once and reused, so that every arm sees identical cells.

Task characterisation (streamed sample)	
conditions characterised	
of which statistically active against DMSO	3,647 / 5
identifiable in principle (ceiling ≥ 0.8)	
with a plate-mate closer than their own replicate	
median cells per condition	
median #DEGs per condition (BH $q < 0.05$)	
median SNR (effect $\div$ replicate noise)	
drugs with a real dose series	
drugs seen in ≥ 3 cell lines	
Residual-frame cache (rebuilt targets, Section 3.7)	
cells	620,564 (600,000 treated)
(cell line, plate) groups	
drugs / cell lines	
conditions with ≥ 40 treated and ≥ 20 control cells	
by split after inventory filters (train / held-out wells / held-out drugs)	
retained by the data-resolved reliability line	2,523 of
training examples / well-held-out validation examples	
Evaluation sets	
tier 1 (seen conditions) / tier 2 (unseen drugs) / tier 3 (unseen combinations)	
conditions scored per tier in the generation evaluations	
tier-2 drugs in the expression-frame NIR benchmark (within plate)	
conditions scored in the residual-frame evaluation	570 (41)

Table 1: The populations behind every n quoted in this thesis. Sources: `drug_biology_atlas.json`, `eval_endcell.cpu.json`, `nir_sameplate.json`, the rebuilt target report and the residual-frame evaluation. Note that the 72.7% active figure is over the 5,014 conditions with enough cells for the permutation test, not over all 6,628.

Three evaluation tiers, defined by what the training set contained, are used for the broad characterisation. *Tier 1* holds conditions whose (drug, cell line) pairing was seen in training, so it measures fit rather than generalisation. *Tier 2* holds conditions whose *drug* never appeared, which is Leave-Drugs-Out in the vocabulary of Section 2.4. *Tier 3* holds pairings whose drug and cell line were each seen but never together — Leave-Pairs-Out. Tier 3 is small (99 conditions) and is reported only where it is adequately powered; the

residual arm builds its own, larger holdout, and the unit that holdout is keyed at turns out to be the most consequential choice in this chapter (Section 3.8).

3.2 Model and representation

A cell is written as a *cell sentence*: its expressed panel genes in descending order of expression, terminated by [END_CELL]. The prompt supplies cell line, drug, dose, mechanism, and the untreated control cell’s own sentence; the response is the treated cell’s sentence. Two tokens, [END_CELL] and — for the residual arm — [DOWN], are registered as atomic special tokens; left unregistered, [DOWN] splits into sub-words and the up/down boundary ceases to be a clean symbol.

The backbone is C2S-Scale-Pythia-1b-pt [3], a C2S-Scale checkpoint built on the Pythia-1b decoder [7], cold-started for every arm from the same base checkpoint with identical hyperparameters (1 epoch, learning rate 10^{-5} , gradient accumulation 16, maximum length 8192, bf16), so that in each experiment the training target or objective is the only variable.

What that checkpoint has already seen bears directly on how the held-out splits should be read. It is pretrained on over 50 million human and mouse cells drawn from 825 single-cell datasets in CELLxGENE and the Human Cell Atlas — an *atlas* corpus, carrying no chemical or genetic perturbation data, and the released model card lists cell and tissue prediction, conditioned generation and gene-set conversion among its capabilities but not perturbation prediction. The base model therefore reaches our fine-tuning having never seen a drug-treated cell, which is what allows the unseen-drug split to mean what it claims: a drug held out here is unseen by the whole pipeline, not merely by its final stage.

3.3 Metrics

The first thing this thesis needs is a ruler that cannot be gamed, because the field’s default one can be — and Section 4.1 exhibits the exploit rather than asserting it. The default is a correlation between predicted and true *change from control*, restricted to the genes that actually move. Writing x for the control profile, y for the treated truth and $\hat{y}$ for the prediction, let

$$S_K = \arg \text{top-}K_g |y_g - x_g| \quad (6)$$

be the K genes with the largest true change. Then

$$\text{DE-}\Delta r = \text{corr}((\hat{y} - x)|_{S_K}, (y - x)|_{S_K}). \quad (7)$$

We report $K = 50$ with Pearson correlation throughout; $K \in \{20, 100, 200\}$ and the Spearman variant were computed and move the numbers by less than 0.02 except at $K = 200$, where the top- K set starts to include genes that do not move.

Three properties of this definition, each innocent in isolation, compose into the failure. First, S_K is selected using the *truth*, so every predictor is scored on the genes the condition actually moved — the exam questions are chosen from the answer key. Second, treated cells in this atlas regress toward a shared response, so a prediction that simply reverts toward the population centre correlates with $y - x$ *by construction*: no drug knowledge is required to score well, only knowledge of where the centre is. Third, the fix is available in the same breath — **partial-DE- Δr** regresses the control profile out of both arguments before correlating, which removes the reversion route and is therefore the honest version of the same quantity. Section 4.1 shows what each version scores a predictor that contains no information at all.

A cell sentence lists only expressed genes, so a predicted profile leaves some panel genes unranked, and any rank-based metric needs a convention for the absent tail. Two were used throughout and reported side by side: **worst** assigns every absent gene the last rank P , and **francesca** assigns them a fixed middle rank $P/2$. The choice is immaterial — spike-in DE- Δr reads 0.952 against 0.954 under the two, and no baseline or ceiling in this thesis moves by more than 0.01 — so all figures are quoted under **worst** and the convention is not mentioned again. For a subset-free alternative we also compute panel- τ , Kendall’s τ between predicted and true rank over the full 946-gene panel, which is invariant to the absent-gene convention by construction.

The metric this thesis relies on is the Normalised Inverse Rank, introduced by Wu et al. in PerturbBench [16] (there called the “transposed-rank” metric) and identified by Miller et al. [14] as one of the few consistently calibrated metrics across 14 Perturb-seq datasets. For a prediction $\hat{y}$ of condition i , with truths $\{y_j\}$ over the comparison set,

$$\text{NIR}(\hat{y}, i) = \frac{1}{|J|} \sum_{j \in J} \left[\mathbf{1}\{\sigma(\hat{y}, y_i) > \sigma(\hat{y}, y_j)\} + \frac{1}{2} \mathbf{1}\{\sigma(\hat{y}, y_i) = \sigma(\hat{y}, y_j)\} \right], \quad (8)$$

the fraction of *other* conditions whose truth the prediction resembles less than its own, with ties counted at one half. Two properties do work that no other candidate does. Chance is 0.50 *by construction*, not empirically: “this model is at chance” is a statement, not a comparison against a fitted null, and that is what makes the negative results in this thesis quotable without a control arm. And because the generic response is present in every condition’s truth in equal measure, it cancels in the comparison — NIR is a *discrimination* metric and only the drug-specific difference decides it, which is precisely the component the reversion artifact exploits. The tie term is not cosmetic: in residual space (Section 3.7) a predictor of “the average drug response” is exactly the zero vector, every similarity against it is zero, and a strict inequality scores it 0.000 rather than the 0.500 it deserves.

Which NIR variant, and why it must be stated. Our instruments compute NIR under two similarity functions and store both: `nir_expr`, Euclidean distance in expression space, and `nir_rank`, the same construction over rank vectors. **Every NIR figure in this thesis is `nir_expr`**, and the choice is load-bearing rather than incidental: under `nir_rank` the tier-2 aggregate places *every* arm below chance, including the noise ceiling itself (0.380, against the model’s 0.313 and a drug-agnostic linear map’s 0.285). A metric on which a real replicate of a condition fails to be retrieved above chance is not measuring what its name suggests, and it is not the metric the calibration analysis of Section 4.1 endorsed — the DRF result is computed on the expression-space variant. We report the calibrated one and name it here so the stored artifacts cannot be misread.

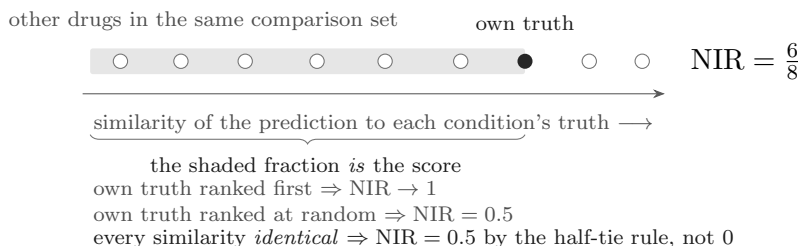


Figure 1: **NIR asks whether a prediction resembles its own condition’s truth more than it resembles the truths of the other drugs in the same comparison set.** The generic response — the stress and cell-cycle program every compound triggers — is present in every truth in equal measure, so it cancels in the comparison and only the drug-specific difference decides. Two consequences follow, and both matter downstream. Chance at 0.50 is a statement about the *comparison set*, not about the model: conditions are grouped by (cell line, plate) in the expression frame and by cell line in the residual frame, and the two are not interchangeable. And a predictor whose similarities are all identical — the zero vector, which is what “predict the average drug response” becomes in residual space — must score 0.50 rather than 0, which is why ties are counted at one half rather than dropped.

Finally, the metrics themselves need grading, and for that we port the framework of Miller et al. [14], asking how well each *metric* separates a positive from a negative control:

$$\text{DRF}(m) = \frac{m(\text{pos}) - m(\text{neg})}{m(\text{perfect}) - m(\text{neg})}, \quad (9)$$

with pos an interpolated-duplicate noise ceiling — their positive control, which interpolates a technical duplicate toward the mean baseline with per-gene weights given by differential-expression significance — and neg a drug-agnostic leave-one-out mean. A negative DRF means the metric scores the uninformative baseline *above* the noise ceiling: it is inverted, and rewards the generic gene ordering rather than the drug. Three hardening choices matter for how the result can be read. Tied expression values receive *average* ranks, because breaking ties by array index makes a tied gene’s rank depend on its position in the panel. The DRF ratio is recomputed *whole* inside every bootstrap resample,

because its denominator is estimated from the same data as its numerator and holding it fixed understates the interval. And the family of five metrics is controlled simultaneously with Holm’s procedure, because the claim “only NIR is calibrated” is a claim about all five at once.

Adopting this framework is deliberate. Miller et al. use it to argue that reports of deep models failing to beat uninformative baselines reflect miscalibrated evaluation rather than model failure (Section 2.3). Evaluating our model on the metric *their* analysis endorses means the result reported in Section 4.2 cannot be dismissed on the grounds their own paper raises.

3.4 Controls

The negative results in this thesis are only as strong as the controls that make them interpretable. Each of the following exists because a specific objection would otherwise be fatal, and each is stated with the failure it closes.

Drug and plate are partially confounded by Tahoe’s design: each drug is assayed on its own plate(s), with roughly 4–12 drugs per plate and plate-matched controls. Discrimination grouped by cell line alone therefore lets the *batch* signature stand in for drug identity. Measured directly, by scoring the same tier-2 conditions with and without the plate held constant:

Predictor (drug-agnostic)	cross-plate	within plate
mean baseline	0.399	0.161
control-copy	0.551	0.510
noise ceiling	0.625	0.575
fraction of drugs at chance	43%	49%

Table 2: Expression-frame NIR. A drug-agnostic mean baseline more than doubles its score when the plate is free to vary, which is what plate leakage looks like. The ceiling falls once the plate is held constant because the batch signature was inflating it too.

The mean baseline is the clearest demonstration: it contains no drug information whatsoever, and it scores 0.399 when the comparison set may span plates against 0.161 when it may not. Every discrimination instrument therefore gained a `--same_plate_only` mode and all discrimination results in this thesis were re-run within plate. Conclusions are unchanged; some magnitudes shrink.

A separate and stronger observation, which belongs to the drug-blindness argument rather than to this one, is that *within* plate and on the identifiable stratum a control-copy baseline reaches 0.766 against the model’s 0.768 (Section 4.2). That is not a statement about plate leakage — it is a statement about the model.

The scramble ablation, and what its comparator must satisfy. The central manipulation leaves the prompt byte-identical except that the drug name and mechanism are replaced by another drug’s, while the control cell and the scoring truth are unchanged. Because both arms share the same control cell, control- and batch-derived leakage cancels in the *difference* $\Delta_{\text{scr}} = \text{NIR}_{\text{model}} - \text{NIR}_{\text{scramble}}$. This is the only manipulation that isolates drug use, and every drug-use claim in this thesis is anchored to it rather than to a comparison against a drug-agnostic predictor.

Two constraints on the swap partner are required for a gap measured this way to mean what it says, and both exist because the unconstrained pool fails in a specific way. The partner must be a *different drug*: allowing the same drug at another dose or plate substitutes the drug for itself, an unchanged output is then correct rather than blind, and the contamination concentrates in the most similar stratum because a drug’s own other doses are usually its nearest neighbours. And the partner must be on the *same plate*: residuals retain plate structure, so a cross-plate partner differs from the target in batch as well as in drug, and the gap absorbs both.

A scramble that swaps in a drug with a *similar* true response is also a weak test, and this is a real risk here because same-mechanism drugs are barely more similar than random pairs (within-MoA distance 2.322 versus between-MoA 2.377, ratio 0.977), so “different mechanism” does not imply “different response”. Two remedies are used: a *swap-distance sweep* that turns the single test into a curve of Δ_{scr} against the response distance $\|y_A - y_B\|$ between the real and swapped-in drug; and a *stratified scramble* with three defined partners per condition chosen by residual cosine within the cell line — **near** (most similar), **orth** ($\cos \approx 0$) and **opposite** (most anti-correlated). The strata make an assumption visible that a single comparator hides: the gap has two parts, how much naming the truth helps and how much naming a lie hurts, and only the first is wanted. A usable null is therefore a stratum whose own score sits at chance — which, Section 4.7 shows, is true of exactly one of the three, and not the one an evaluation would naturally default to.

The scramble replaces the drug name *and* its mechanism together, so a non-null gap establishes sensitivity to the combined drug-and-mechanism prompt without attributing it to either span. A *field decomposition* costs one extra pair of generation arms and swaps exactly one span of the instruction line to the opposite-signature partner, leaving every other byte identical: the drug name with the mechanism kept, the mechanism with the name kept, and both together. One guard is essential rather than tidy. If the swap partner happens to share the original’s mechanism, then swapping only the mechanism produces a prompt *identical* to the model’s own, and the arm scores a gap of exactly zero *by construction*. Read naively that is indistinguishable from “the model ignores the mechanism”, which is the very claim under test. Such conditions are dropped rather than scored, which is why the mechanism arm carries fewer conditions than the others.

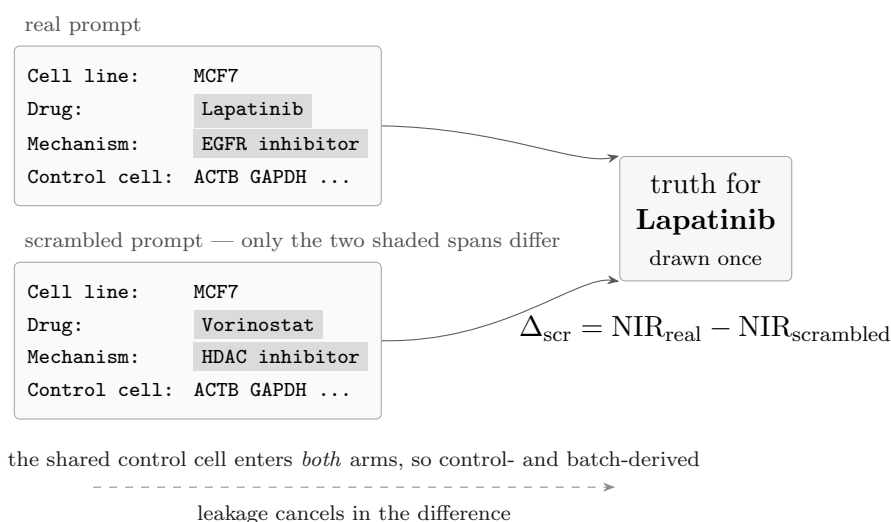


Figure 2: **The scramble replaces the drug name and its mechanism and nothing else.** Both arms are conditioned on the same control cell and scored against the same truth, so any leakage through the control cell or the plate enters both arms in equal measure and cancels in the difference Δ_{scr} . This is why Δ_{scr} , rather than a comparison against a drug-agnostic predictor, is the quantity every drug-use claim in this thesis is anchored to: the drug-agnostic comparison is not leak-immune, and a drug-agnostic mean baseline scores $\text{NIR} = 0.399$ when the comparison set may span plates against 0.161 when it may not, precisely because it inherits that leakage. A refinement matters here and is developed in Section 3.4: if the swapped-in drug happens to have a *similar* true response then an unchanged output is correct rather than blind, so the swap partner is chosen from three defined similarity strata and the evidence is the *gradient* across them rather than any single value.

Two further drug-isolating instruments. The drug-blindness claim rests on four instruments rather than one, because they fail in different ways and agreement across them is worth more than any single number. Two are defined above (NIR and the scramble); the other two are forced-choice grading and output invariance.

In *forced-choice grading*, a prediction is scored against its own condition’s truth and against another drug’s truth from the same comparison set, and the instrument records how often the own truth wins. Chance is 0.50 by construction. The ceiling is the same forced choice with a real held-out replicate substituted for the prediction, which bounds what any predictor could achieve at this noise level; it lands between 0.67 and 0.83 depending on cells per condition.

Output invariance asks whether the drug token changes the generated text at all, without reference to any truth. For a fixed control cell, the model generates twice from the *same* prompt and once from each of several prompts differing only in the drug, and the instrument compares

$$\text{gap} = \underbrace{\overline{\text{sim}}(\text{same prompt, resampled})}_{\text{sampling noise alone}} - \underbrace{\overline{\text{sim}}(\text{prompts differing in the drug})}_{\text{noise plus any drug effect}}, \quad (10)$$

with similarity measured as top-100 Kendall τ and as Jaccard overlap. A positive gap means two generations for the same drug resemble each other more than generations for different drugs do, i.e. the drug token moves the output. A gap of zero means swapping the drug perturbs the output no more than resampling at the same temperature does. Reported over 8 contexts $\times$ 12 drugs at temperature 0.8. This instrument is the weakest of the four — it can only detect whether the output moves, not whether it moves *correctly* — but it is also the one least able to be confounded, since it needs no ground truth at all.

Channel gate. For an unseen drug, prediction is only possible through a property the drug *shares* with drugs seen in training, so the reachable channels can be enumerated and each one measured in the same frame. For a channel X , a drug’s residual is predicted as the mean residual of the *other* drugs in the same cell line that share property X with it — the construction of the mechanism baseline, generalised. Because it uses only drugs that were in training, it is a fair contest in every split, unlike a same-drug lookup, which on an unseen drug is an oracle.

The gate’s estimand requires the channel arm and its null to differ in exactly one respect, and two disciplines enforce it. Every channel is paired with a **plate-matched random null**: the same *number* of partner drugs, drawn at random from the same cell line, *and* the same split of partners across the target’s own plate and other plates, because mechanism- and target-related drugs are co-plated more often than chance — a null matched on count alone would differ from the channel arm in plate membership as well as in the channel, which is two differences where the estimand allows one. A

co-plating diagnostic is printed before any verdict, so the size of the correction is a number rather than an argument, and where a plate-matched null spans too few cell lines the channel is reported UNTESTABLE rather than closed — “we could not build the control” must never be written as “closed”. Coverage is likewise reported before any verdict, because a channel annotated on few drugs cannot be closed by a null result: it was never tested.

The bound this establishes is on *similarity transfer* through each channel, not on any model. A model could in principle combine channels or learn a non-smooth structure-to-response map that a nearest-neighbour average cannot express. A leave-one-drug-out regression over all channels jointly addresses the first. The second is bounded by sample size rather than by argument: at a few hundred drugs, learning a non-smooth map is not feasible, so at this n the lookup is effectively the bound.

Is the interaction learnable? The transfer coefficient says how *large* the drug $\times$ cell-line interaction is; it says nothing about whether anything could learn it. If a cell line modulated drug responses in a consistent direction, that direction would be estimable from the line’s own cells and a conditional model would have a concrete target. Writing $\kappa(d, c) = r(d, c) - \hat{\beta}(d)$, the test compares $\cos(\kappa(d, c), \kappa(d', c))$ for different drugs in the *same* cell line against the same quantity across *different* cell lines.

$\hat{\beta}$ must be estimated **leave-one-cell-line-out**. Estimated over all lines it would contain $r(d, c)$ itself, and subtracting a mean containing its own term induces a $-1/(m - 1)$ correlation that biases the signal arm downward *by construction* — which would make a structured interaction look idiosyncratic, the exact error the test exists to avoid. Both arms are disattenuated by κ ’s own split-half reliability, since κ is a difference of two noisy quantities and is therefore noisier than the residual it comes from.

Uncertainty. Conditions are not independent, and they are not independent in two crossed ways at once. Two conditions from the same treated well share a physical assignment and a batch — one well holds about 49 cell lines dosed together — and two conditions from the same cell line share the biology. Treating conditions as independent gives intervals that are too narrow; here, about 1.3 to 1.5 times too narrow. Neither grouping alone captures the dependence, and the tempting shortcut is actively misleading: clustering on the assignment unit alone uses the *larger* cluster count (289 wells against ≈ 50 lines) and would *tighten* every interval while sounding more rigorous. All intervals in this thesis are therefore two-way cluster-robust (Cameron–Gelbach–Miller),

$$V = V_{\text{line}} + V_{\text{well}} - V_{\text{intersection}}, \quad (11)$$

where the intersection of the two groupings is the single condition, so $V_{\text{intersection}}$ is the ordinary independent-sampling variance; when the estimator goes non-positive in a finite sample the wider one-way variance is used and the branch recorded. Because a cluster-robust variance is asymptotic in the number of *clusters*, not observations, the reference distribution is Student’s t with degrees of freedom one less than the smaller cluster count — which matters where an arm spans only 25 wells, and moves a borderline verdict. For statistics computed over *pairs* of conditions, such as the transfer coefficient, neither grouping partitions the observations — a pair belongs to two nodes at once — so those intervals use the Fafchamps–Gubert dyadic estimator, under which two pairs are dependent whenever they share either member. This is the structure DrEval [13] identifies in this field more broadly, citing Hurlbert’s pseudoreplication [17]: hundreds of thousands of cells resolving to a few hundred conditions over 50 cell lines.

Generation validity. Every generation-based evaluation reports the [END_CELL] / [DOWN] completion rate, the fraction of emitted symbols that are valid panel genes, and the duplicate rate, before any score is quoted. This is not hygiene: a 600-token generation cap silently truncated 26% of generations in an early run and *halved* the measured effect. Every contrast is additionally generated with a fixed per-(condition, arm) seed, so the two arms of a paired comparison are not drawn from different points of one global stream.

3.5 Mechanistic probing

To distinguish “cannot represent the drug” from “represents but does not use it”, four instruments are needed, and again each exists because its predecessor fails in an instructive way.

Per-layer *linear probes* on the residual stream at the last prompt token, predicting drug identity, establish whether the information is present at all. But the raw drug subspace is partly a batch subspace — probing dose and plate from *inside* it recovers dose at 0.90–0.95 — so a context subspace is built from the union of cell line, plate and dose, and the drug axis is orthogonalised against it; every causal claim uses the purified subspace. Before any “ablating the drug does nothing” result is trusted, a *removal gate* requires the subspace to contain the drug: the fraction of above-chance probe accuracy removed by projecting it out must exceed 0.8, without which the null is unfalsifiable. The *causal test* then reads log-probabilities of the teacher-forced true response clean, then re-reads them with a forward hook projecting the subspace out at every position, taking the mean KL divergence over response tokens; measuring in logit space rather than by generating removes the sampling-noise floor that made an earlier steering experiment inconclusive.

The identity test, and two designs that failed before it. Ablation shows a subspace carries functional variance; it cannot show that the subspace’s *drug-specific content*

is read. Only a substitution can. Two attempts at one were retracted, and since the failures were instructive they are stated rather than hidden.

The first overwrote drug A’s code with drug B’s inside a purified subspace and compared the resulting KL against a random vector of identical norm drawn in the *full* residual stream — so almost all of the control’s energy lay in directions the intervention never touched. The second drew the random vector inside the subspace, which appeared to fix it and did not: the hook ablates before adding, so the delivered perturbation is (added – removed), and because the injected class mean was uncentred it shared its global-mean component with the thing removed. The substitution arm was then a near-restore and the control arm a genuine displacement. Simulated with no model in the loop, the control delivers $5.3\times$ the perturbation in 100% of draws. Both retracted results pointed the same way, and that way was geometry rather than biology.

The design used here removes the ablation from the identity test. It injects a *displacement* $P(\mu_B - \mu_A)$ — in which the global mean cancels exactly — and scores a *contrast*,

$$\log P(r_B \mid \text{prompt}_A) - \log P(r_A \mid \text{prompt}_A), \quad (12)$$

so that any shift common to both responses, including the entire normaliser, subtracts out. What survives is the model’s preference between drug B’s own held-out response and drug A’s. The headline control is a second real drug displacement $P(\mu_C - \mu_A)$, rescaled to identical norm: same anchor, same construction, wrong drug. An isotropic in-subspace direction is retained only as an off-manifold damage diagnostic, never as the comparison, because a random direction disrupts the model in ways an on-manifold class-mean displacement does not.

Subspaces and class means are fit on one split, r_B drawn from a second disjoint split, and all scoring done on a third. Uncertainty is a cluster bootstrap over (A,B) pairs; equivalence requires the interval to lie inside a pre-registered margin *and* contain zero; and a layer is excluded where the cell-line positive control has no dynamic range, since a saturated instrument cannot support either verdict.

3.6 Target constructions

The three target families below are attempts to fix the target *content* while leaving its encoding alone; Section 4.4 shows why that family of fix could not work, and the ladder construction is what makes that demonstration conclusive rather than anecdotal.

Consensus. Cells grouped by (drug, cell line, dose) and averaged into one pseudobulk sentence, with every original prompt retained so only the response changes.

Optimal transport. For each (cell line, plate) group, a coupling π between control and treated cells minimising $\sum_{ij} \pi_{ij} \|x_i - y_j\|^2$ under uniform marginals, solved by Sinkhorn iteration [24] on $K = \exp(-C/\varepsilon)$. Couplings are computed in the released Tahoe scVI

latent [26] ($n_{\text{latent}} = 10$), joined per cell by barcode rather than re-encoded — a re-encode produced a degraded latent (cell-line probe 0.11 versus 0.975 for the shipped latent). Targets are built *decoder-free*, as convex combinations of real treated cells in expression space, so they always lie on the data manifold.

The regularisation parameter defines a ladder whose endpoints are experiments we had already run: $\varepsilon \rightarrow \infty$ gives a uniform coupling and hence the treated mean (consensus); $\varepsilon \rightarrow 0$ gives near-hard assignment, approximately a single state-matched treated cell. This is what makes the sweep conclusive rather than anecdotal.

3.7 Residual targets

The residual target asks the model to predict only what makes *this* drug different. For each condition (drug, cell line, plate, dose) with at least 40 treated and 20 plate-matched control cells:

$$r(d, c) = \underbrace{(\bar{y}_{\text{treated}} - \bar{y}_{\text{control}})}_{s(d, c)} - \underbrace{\frac{1}{|D_c|} \sum_{d' \in D_c} s(d', c)}_{\text{generic}(c)}. \quad (13)$$

The control subtraction removes the cell’s own state; the mean-over-drugs subtraction removes the *generic drug program* — the stress and cell-cycle response common to all compounds, which Section 4.1 shows accounts for essentially all of the model’s apparent skill. What remains is what makes *this* drug different. Four properties of how the generic is estimated decide whether that remainder can be trusted, and each was forced by a measured failure.

First, **the split precedes the fit**. An earlier build estimated the generic and the reliability filter over *every* condition and assigned the holdout afterwards, so each held-out target was defined partly by other held-out conditions. The pipeline is therefore staged — inventory (metadata only), split (metadata only, before any pseudobulk exists), fit (the generic is handed the training keys explicitly), transform (held-out conditions are projected through the fitted generic and never enter it) — and a poison test stands behind the staging: corrupting every held-out expression vector must leave the training targets byte-identical, and corrupting a training condition must change them.

Second, **the scope is the plate, not the cell line**. The original argument for cell-line scope was reproducibility — 62% of conditions reproduce across a half-split at cell-line scope against 19% at plate scope, on the reasoning that a plate holds too few drugs to estimate a mean and subtracting a noisy one injects noise. That comparison does not hold up: both figures were computed with a *shared* control pseudobulk subtracted from both halves, which makes the halves correlate through their common control error, and the inflation is not neutral between scopes — it favours cell-line scope precisely because the cell-line generic is itself less noisy. Measured with disjoint control halves, the two are

comparable: 20% at plate scope against 16% at cell-line scope. Two further diagnostics settle it in plate’s favour and are reported in Section 4.5: at cell-line scope the dose-transfer ordering inverts and the shared-minus-split control bias reads +0.254, against −0.000 at plate scope. A plate group with fewer than three other training drugs yields no generic, and the condition is dropped and counted rather than silently falling back to the contaminated frame.

Third, **the mean is over drugs, not conditions, and leaves the condition’s own drug out**. A drug measured at five doses would otherwise contribute five times the weight of a single-dose drug. And once the generic is train-only, an asymmetry appears that had not existed before: a training condition’s generic would contain its own drug while a held-out condition’s would not, so the two splits would be scored against differently defined targets and the generalisation gap would partly measure that difference. The generic for every condition is therefore “the mean over training drugs other than this one, within scope” — identical in definition on both sides.

Fourth, **the reliability line is resolved from the data, not inherited**. A condition is kept for training only if its residual reproduces across a half-split, $\cos(r_A, r_B)$ above a threshold — with each half measured against its *own* control half, so the two share nothing. The threshold itself was the single largest determinant of the training set, and it had been inherited (0.2) from an era when reliability was measured with shared controls — a different quantity on a different scale, under which the mean is ≈ 0.6 rather than +0.127. Rather than pick a new number, the builder reports the whole retention curve against a null that pairs half A of one condition with half B of a *different* condition — same encoding, same dimensionality, same noise, no shared biology — and the threshold is set at the null’s 95th percentile: a condition is kept when its two halves agree more than two unrelated conditions’ halves do. At the measured null (95th percentile +0.109) this retains 2,523 of 4,999 training conditions (50.5%). The held-out sets are *not* filtered by this criterion: dropping held-out conditions for failing their own reproducibility bar would select the evaluation set on its own outcome, so noise in retained training conditions costs learning efficiency, not validity.

What this fraction means is stated carefully, because it is easy to over-read. The half-split agreement is a *within-well sampling-precision* criterion: it asks whether two disjoint cell subsets of the *same* well give the same residual. It is not biological reproducibility — 286 of 287 distinct (drug, dose) treatments in this atlas occur in exactly one well, so no independent-well replicate exists to measure that with — and the 50.5% is a statement about how much drug-specific signal survives pseudobulk noise at this cell budget, not about how much biology is real.

Encoding and scoring. The target string is <up genes> [DOWN] <down genes> [END_CELL], 100 genes per block, ordered by signed residual magnitude; sign is biology, so it is given a

symbol rather than being folded into a magnitude ranking. Predicted and true residuals are both mapped to a signed rank vector (top- k up positive, top- k down negative, weight $1/\log_2(\text{rank} + 1)$) so a generated sentence and a continuous truth are compared like for like; similarity is cosine and the score is NIR over other drugs in the same cell line.

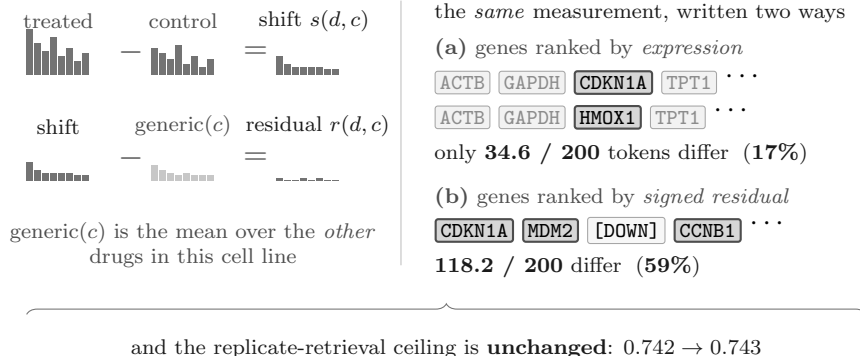


Figure 3: **The information was always present; the tokenisation discarded it.** Left: the drug-specific residual is built by two subtractions. The control removes the cell’s own state, leaving the *shift*; the mean over the other drugs in the same cell line removes the *generic* program that every compound triggers, leaving what makes this drug different. Right: that same measurement, written two ways. Ranking genes by expression (a) puts housekeeping genes first, so only 34.6 of the top 200 tokens differ between two drugs in the same context and cross-entropy has almost nothing to learn drug identity from; ranking by signed residual (b) raises that to 118.2. The bracket is the point of the figure — the replicate-retrieval ceiling is essentially identical under both encodings, so no information has been added. What changed is how much of it the token sequence exposes to the loss.

3.8 Generalisation design

This is the choice on which every generalisation claim in the thesis stands or falls, and it is forced by the physical design of the atlas rather than by statistics. Tahoe applies one drug at one dose to one well containing ≈ 49 cell lines simultaneously — measured: 289 treatment samples, a mean of 48.6 cell lines per well (median 49, range 43–50), and 286 of 287 distinct (drug, dose) treatments occurring in exactly one well. Three consequences follow, and they apply to every analysis of this atlas, not only this one.

A drug cannot be present in cell line A and absent from cell line B at the same dose, because the two lines are physically the same well. So a (drug, cell line) holdout — the field’s standard Leave-Pairs-Out split, and the split an earlier version of this work used — leaves the held-out condition’s treated well sitting in the training set through its other ≈ 48 cell lines. Measured against the holdout this thesis previously used, 124 of the 268 treated wells in this cache placed a held-out condition and a training condition in the same physical sample. What that estimand measures is *within-atlas matrix completion*, not independent generalisation, and no processing of the targets changes it, because it is a property of what was randomised.

Holding out *whole wells* is the only assignment-respecting choice the design offers, and it answers a different and narrower question: generalisation to an *unseen treatment well of a seen drug* — a new physical preparation, not a new cellular context. Fixed-dose cross-context transfer, present in one line and absent in another, is structurally unidentifiable in this atlas; this is a property of Tahoe’s design, not of our pipeline, and no rebuild fixes it. The split used throughout the residual arm is therefore keyed at the well:

train every condition of the remaining wells (4,999 conditions before the reliability line, 2,523 after it).

unseen_combo — **unseen wells of seen drugs** 36 treated wells held out whole (904 conditions, 900 after inventory filters). The holdout never takes a drug’s last training well — without that guard a two-well drug could silently become an unseen *drug* while sitting in the unseen-combination arm, making the two arms measure the same thing. Zero wells contribute conditions to both sides of this split.

unseen_drug — **Leave-Drugs-Out** every condition of 10 held-out drugs (714 conditions, 711 after inventory filters). The model never sees the token. This is the *control*: naming a drug the model knows nothing about versus naming something else can make no difference, so the measured effect there must be zero — a non-zero value is an instrument fault, not a biological finding.

We do not build a Leave-Cell-Lines-Out (LCO) split here; Section 5 argues it is the natural next axis and explains why its interest is contingent on the transfer coefficient.

A warning we inherit with the vocabulary. DrEval reports, citing [18], that apparently good performance on pair-holdout splits can be reached simply by predicting each drug’s average response across the training cell lines. Our held-out-well estimand sits exactly in the regime where that warning bites: the drug is seen, and its average is computable. This is the reason the baseline ladder below is not optional: the drug-average predictor is not a formality to be dismissed, it is the hypothesis under test.

Stratified sampling. Evaluation samples conditions under an explicit per-split quota. A uniform draw reproduces the pool’s proportions, which starves precisely the split the experiment exists to test: in a first run, 250 constructed held-out combinations yielded only 33 scored conditions, purely as an artifact of the sampler. The residual-frame evaluation therefore scores 200 train, 250 **unseen_combo** and 120 **unseen_drug** conditions — the latter two being quotas of the 900 and 711 available, so the held-out verdicts in Section 4.7 are powered on a subset of the conditions the split affords, and their detection limits should be read with that in mind.

3.9 Baselines

Each baseline is a predictor of the drug-specific residual, scored through an identical pipeline — same truths, same signed-rank encoding, same comparison set.

Baseline	Information used	Role
<code>drug_lookup</code>	this drug’s mean residual, all other cell lines	the bar to beat
<code>drug_lookup_1</code>	this drug in <i>one</i> other cell line	removes the averaging advantage
<code>moa_lookup</code>	other drugs sharing this drug’s MoA, same cell line	mechanism-leak diagnostic
<code>control_copy</code>	none (predict the control cell)	must be ≈ 0.50
<code>generic</code>	none (predict the average response)	must be ≈ 0.50
<code>random</code>	none	floor

Table 3: The baseline ladder. `drug_lookup` is the pre-registered win condition: a lookup can reproduce a memorised average, so only a model that tailors the response to the control cell’s state can beat it.

Two disciplines make the ladder fair. The lookups are fitted from **training conditions only**: fitted from the full cache, a lookup scoring a held-out condition reads residuals the model was never shown — an oracle, not a baseline — and the oracle variants are reported alongside, explicitly labelled, as a bound on how much the comparison was ever inflated rather than hidden. And `drug_lookup_1` selects a cell *line* and averages that line’s conditions rather than selecting a single condition, because 62% of drugs are multi-dose and a single condition would confound “different cell line” with “different dose”. `moa_lookup` is not an oracle in any split, as it uses only drugs that were in training.

3.10 Variance decomposition

To separate the per-drug main effect from the drug×cell-line interaction we model

$$r(d, c) = \beta(d) + \kappa(d, c) + \epsilon, \quad \mathcal{T} = \frac{\text{var } \beta}{\text{var } \beta + \text{var } \kappa}, \quad (14)$$

where β is constant across cell lines and κ is the interaction — the only component a conditional model can win on.

This is DrEval’s naive predictor, in transcriptomic space. DrEval’s `NaiveMeanEffectsPredictor` predicts $\hat{y}_{ij} = \mu + (\mu_i^c - \mu) + (\mu_j^d - \mu) = \mu_i^c + \mu_j^d - \mu$: an additive model of cell-line and drug main effects, with no interaction term. Our `generic(c) +  $\beta(d)$` is precisely that quantity, lifted from a scalar potency to a 946-dimensional profile. It follows that κ is the term their naive model omits, and that $1 - \mathcal{T}$ measures exactly the headroom available above it. DrEval finds that deep models barely beat this additive baseline on drug sensitivity; the transfer coefficient asks whether, in the transcriptomic setting, there is anything above it to win in the first place.

$\mathcal{T}$ is estimated by disattenuated cross-line correlation,

$$\hat{\mathcal{T}} = \frac{\cos(r(d, c_1), r(d, c_2))}{\sqrt{\rho(d, c_1) \rho(d, c_2)}}, \quad \rho = \frac{2 \cos(r_A, r_B)}{1 + \cos(r_A, r_B)}, \quad (15)$$

with ρ the Spearman–Brown reliability of the full-sample residual. Dividing out the reliability is essential — measurement noise alone depresses the raw cross-line cosine and would otherwise be read as interaction — and the reliability entering it is a *within-well* split-half precision, which is optimistic relative to true biological reproducibility. That optimism sits in the denominator, so the interaction share $1 - \mathcal{T}$ is, if anything, understated.

Five controls accompany the estimate, each capable of changing the conclusion:

1. **Control half-split.** Subtracting one shared control pseudobulk from both halves leaves control-estimation error common to r_A and r_B , inflating the reliability and biasing $\mathcal{T}$ *downward* — manufacturing apparent interaction. Both builds are run and the bias reported (+0.254 at cell-line scope, −0.000 at plate scope).
2. **Negative control, structure-matched.** A different drug in a *different* cell line, breaking only the drug match while preserving the cross-line structure of the estimand; it must read ≈ 0 . A same-plate/different-drug control was tried first and is *not* adequate: it differs from the estimand in two ways at once, and plate structure surviving a cell-line-scoped generic inflates it to +0.478 while leaving the estimand untouched — which produced a false-alarm rejection before the matched control was built.
3. **Simulated null.** At $\kappa = 0$ the estimator does not return 1.0. A simulation calibrated to the measured noise level fixes the reference. In validation the raw cross-line cosine at $\kappa = 0$ is 0.734, which read naively reports 27% interaction where the truth is zero.
4. **Dose strata.** Same drug, same cell line, different dose is not cross-line transfer; it is reported separately as a reference scale, with dose resolved to a *molar* quantity — an earlier cache carried the well identifier in the dose field, so what was labelled dose transfer compared physical samples rather than concentrations.
5. **Leave-one-out generic.** The generic includes drug d , leaving a $-1/m$ self-term, unless the condition’s own drug is excluded — which, per Section 3.7, it now is.

Validation against planted ground truth recovers $\mathcal{T} = 1.00, 0.70$ and 0.40 to within 0.005 , and the different-drug negative control returns -0.001 .

Finally, an exact energy decomposition of the shift into residual, generic and their cross term — the three shares sum to one by construction — reports what fraction of the

total perturbation response the drug-specific frame covers: the number that scopes every claim in this thesis, and one that has not previously been computed for this task.

4 Results and Analysis

This chapter makes one argument in five moves, and the order is load-bearing. The field’s standard metric is shown to be gameable before any model is scored, because every later claim depends on the ruler. Under a ruler that cannot be gamed, the model is drug-blind, and the blindness is located in what the model is asked to predict rather than in what it can represent. Re-encoding the prediction target then produces genuine prompt sensitivity — and the chapter’s central result is that this sensitivity does not become reliable held-out prediction: a training-only lookup table beats the model decisively, and the gap between the two is where the thesis ends.

4.1 The metric is the first result

Before asking what the model knows, we have to know what the scoreboard rewards. The answer, for the field’s default metric, is: nothing about the drug. A predictor that places every gene at the middle rank — carrying no information about drug, cell or biology — scores $\text{DE-}\Delta r(K=50) = 0.9999$ [0.99991, 0.99992], above the two-real-cell noise ceiling (0.76) and above the model (0.73). The ordering is inverted: less information yields a higher score, which is the signature of an artefact rather than of skill. The same predictor’s partial-DE- Δr and panel- τ are *undefined* rather than merely poor — its shift from control is identically zero, so there is nothing left to correlate once the control is regressed out. The cause is exactly the mechanism Section 3.3 flagged: DE- Δr selects the scored genes using the truth, and a prediction that reverts toward the centre correlates with the true change by construction.

Predictor	Information	DE- Δr (K50)	partial-DE- Δr
revert_center (all genes $\rightarrow P/2$)	none	0.9999	undefined
revert_mean (predict mean control)	none, no fit	0.961	0.25
ridge linear (control $\rightarrow$ shift)	control	0.947	0.28
noise ceiling (two real cells)	—	0.76	—
model (1B LLM)	control + drug	0.73	not measured

Table 4: DE- Δr is saturated by a control artefact. Regressing the control out of the two fitted baselines leaves ≈ 0.26 of real but *drug-agnostic* skill, and a baseline that performs no fitting at all reaches the same figure. The model’s own partial-DE- Δr was never computed — it requires a GPU pass that was not run — so the cell is marked rather than filled with the expected value; the model’s panel- τ , which *is* measured, carries the same conclusion below.

Removing the control (partial-DE- Δr) collapses the raw 0.95 to ≈ 0.26 for the two baselines that were scored — and that residue is matched by a baseline that performs no fitting at all. The model’s own partial-DE- Δr is not measured, so the comparison is

anchored instead to panel- τ , which is measured for every arm and tells the same story: `revert_mean` 0.268, ridge 0.265, model 0.257 (tier 2, `worst` convention). The response decomposes into control reversion (an artefact), a generic drug program (≈ 0.26 , drug-agnostic, trivially matched), and a drug-specific component that no predictor captures.

Grading the five metrics against each other settles which instrument the rest of the thesis may use. Under the hardened DRF protocol of Section 3.3 — tie-aware ranks, the ratio resampled whole, the family controlled by Holm’s procedure — NIR is the only calibrated metric: DRF = +0.635 [+0.618, +0.652], Holm $p = 0.0025$. The four rivals remain strongly negative and none approaches significance (Holm-adjusted $p = 1.0000$ for all four); in the earlier unhardened calibration they read -0.081 (`weighted_r2`), -0.231 (panel- τ), -0.415 (`spearman_expr`) and -0.448 (DE- Δr), and a negative DRF means the metric scores a drug-agnostic baseline *above* the noise ceiling. The inversion is a property of the data, not a leak: a leave-one-out correction barely moves the numbers, since real cell lines carry 20–204 drugs, and it survives holding the plate constant.

This replicates Miller et al. — and then diverges from them. Their central methodological claim is confirmed emphatically: metric choice dominates, the rank-based metric is the one that survives, and the correlation-based metrics they identify as poorly calibrated are exactly the ones that fail here. The agreement extends to the specific metrics — they find NIR consistently calibrated across 14 Perturb-seq datasets, and we find it is the only calibrated metric of the five we tested on drug perturbation. Our `weighted_r2` result is also consistent with their finding that weighted metrics behave better than unweighted ones. The divergence comes at the next step. Miller et al. conclude that under calibrated metrics deep models outperform uninformative baselines; Section 4.2 evaluates on the calibrated metric and finds the model at chance. Two differences in setting are candidates for the discrepancy — genetic versus drug perturbation, and a discriminative regression model versus an autoregressive generative one — and Section 4.8 supplies a third and more structural possibility.

The metric can discriminate even if the model cannot: a spike-in forced choice between two real drug populations separates them essentially perfectly at pseudobulk (panel- $\tau = 1.000$, DE- $\Delta r = 0.995$, top- n $\tau = 0.987$, plate held constant). The instrument is not the bottleneck.

4.2 The model is drug-blind

Having established which instrument can be trusted, we can finally ask the question the thesis exists to answer. The answer is uniform across four instruments that fail in different ways, which is the only reason to report all four.

Before asking what the model knows, it is worth asking what there is to know. A ladder of mean-shift baselines — each predicting the average response of a progressively narrower

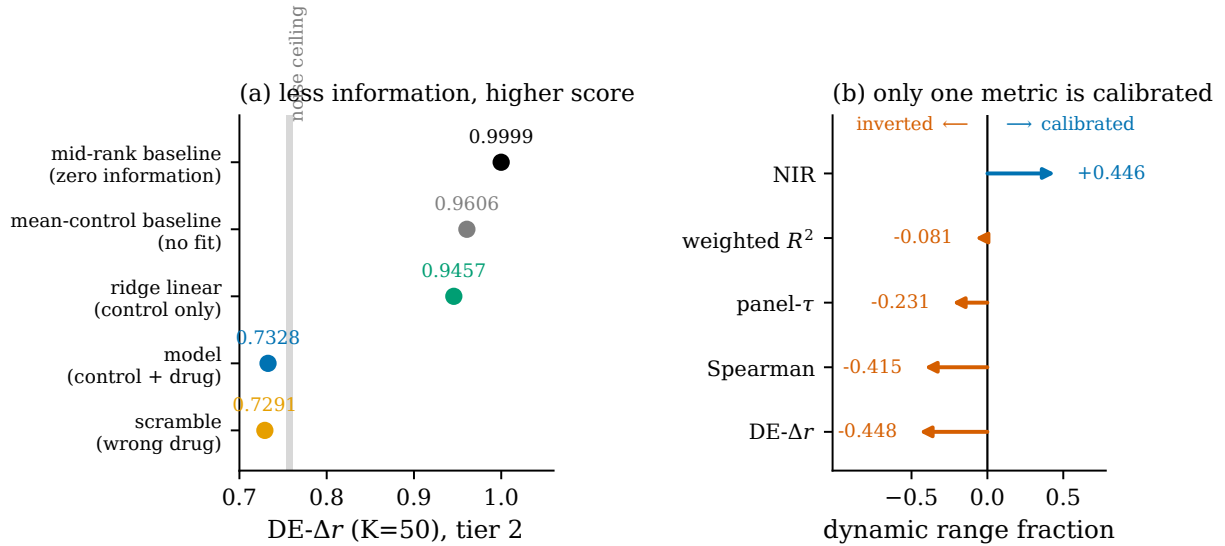


Figure 4: **Under the standard metric, less information yields a higher score.** (a) A predictor that places every gene at the middle rank — carrying nothing about drug, cell or biology — scores $\text{DE-}\Delta r = 0.9999$, above the two-real-cell noise ceiling and above both the model and its own scramble. The inverted ordering is the signature of an artefact rather than of skill, and its cause is control regression-to-the-mean. Tier 2, $n = 400$ conditions. No intervals are drawn: the stored ones are normal approximations rather than the cell-line-clustered bootstraps used everywhere else in this thesis, and mixing the two conventions on one axis would misrepresent both. (b) Grading the *metrics* rather than the models. The displacement drawn is the Dynamic Range Fraction, on a scale where 0 is the drug-agnostic baseline and 1 a perfect predictor, so a negative value means the metric ranks an uninformative baseline *above* the noise ceiling. Only NIR points forward; the hardened estimate is $+0.635$ [$+0.618$, $+0.652$]. 655 drugs over 61 cell-line $\times$ plate groups, within plate. `weighted_r2` carries no directional claim — its sign tracks cell count, and it is approximately neutral.

Instrument	Value	Reading
Forced-choice grading	0.48 ($\approx$ scramble; ceiling 0.67–0.83)	chance
Output invariance (gap)	0.000 [−0.019, +0.016]	null
Scramble DE- Δr , tier 1	0.740 real vs 0.739 scrambled	identical
Scramble DE- Δr , tier 2	0.733 real vs 0.729 scrambled	identical
Δ_{scr} (identifiable subset)	+0.014 [−0.016, +0.042]	null

Table 5: Four independent instruments agree. Near-ceiling DE- Δr is achieved *identically* with a wrong-mechanism drug, so the score is drug-independent.

group — answers that directly. On tier 2, predicting the global mean scores DE- Δr 0.056; conditioning on cell line roughly triples it to 0.142; conditioning on mechanism does not move it (0.052); and conditioning on mechanism *and* cell line is no better than cell line alone (0.120 against 0.142). Tiers 1 and 3 tell the same story. The informative conditioning variable in this data is therefore the cellular context, not the compound class — worth holding onto, because Section 4.3 finds precisely that asymmetry inside the model’s activations, and Section 4.10 finds that mechanism does carry signal once the generic program is subtracted. The apparent tension is a frame difference rather than a contradiction: in full-profile space the generic response dominates and swamps a mechanism effect that becomes visible only in the residual frame.

On the calibrated metric the model sits at chance and is matched by predictors that know nothing about the drug: within plate on tier 2 ($n = 606$), the replicate ceiling is 0.576, the model scores 0.498, a drug-agnostic linear map 0.500, a zero-drug-information control-copy 0.504, and the global mean 0.180.

Two objections to this verdict close on measurement rather than argument. First, *is the task winnable at all?* Only 46.2% of (drug, cell line) conditions are identifiable in principle; 27.3% of drugs are statistically inert against DMSO and 78.7% have a plate-mate closer than their own replicate. But restricting to the identifiable subset does not rescue the model: it appears to beat the linear baseline there (0.768 vs 0.531), yet a zero-drug-information control-copy matches it exactly (0.766) and the leak-immune Δ_{scr} remains null. The model is drug-blind *even where the task is provably winnable*. The ceiling itself rises with aggregation (0.61 at 10 cells, 0.76 at 40, 0.85 at 120), so the signal is real and noise-limited.

Second, *does the representation throw the drug away?* Cell sentences keep only the rank order of genes and discard expression magnitude, and if the drug lived in the magnitudes the diagnosis of the following sections would be attacking the wrong thing. The objection is testable without any model, by asking whether *true* expression discriminates drugs any better than rank does. It does not: at single-cell resolution, cosine, Pearson and Spearman similarities on true normalised expression all sit within 0.04 of chance (0.529–0.531 [0.515, 0.543]) — the same near-chance regime as rank. What recovers the signal is

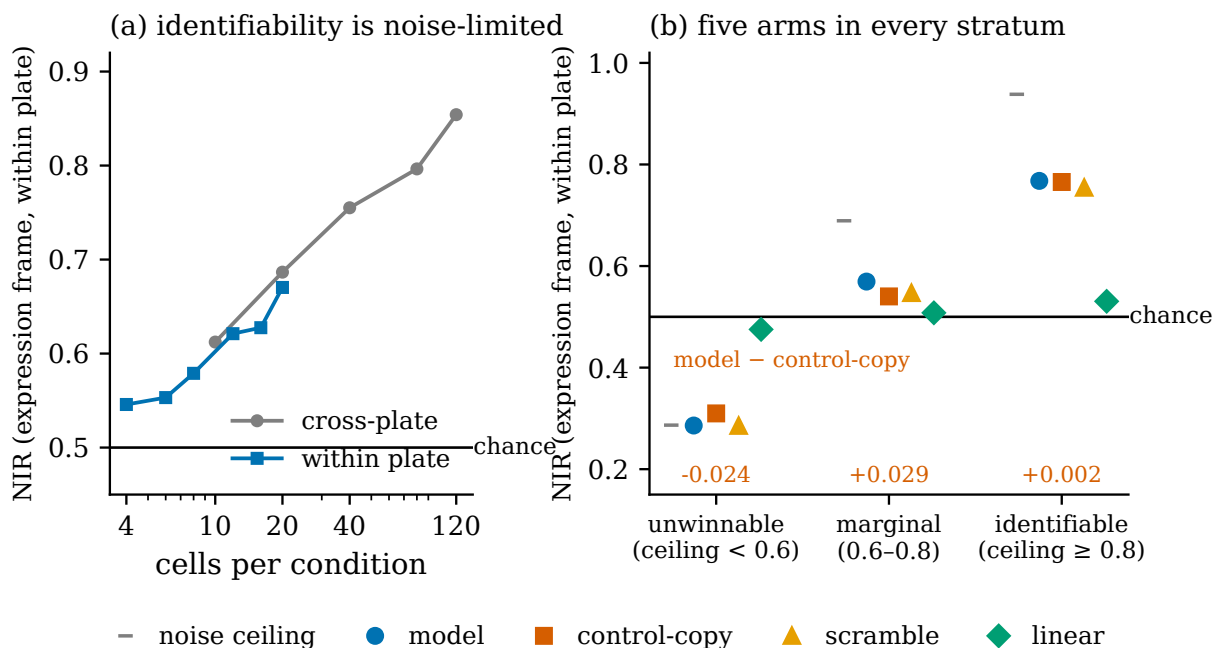


Figure 5: **The task is winnable where the noise ceiling is high — and precisely there, a predictor containing no drug information matches the model.** (a) The ceiling rises steadily with the number of cells per condition, cross-plate and within plate, so identifiability is noise-limited rather than absent: the drug signal is real and emerges under aggregation. (b) Conditions binned by their own ceiling, with five arms in every stratum. On the identifiable stratum ($n = 204$) the model reaches 0.768 — and a control-copy baseline that simply reproduces the untreated cell, and therefore knows nothing whatever about the drug, reaches 0.766. The paired contrast beneath each stratum is $+0.002$ $[-0.026, +0.028]$ there, clustered over cell lines. Showing the model alone across these strata would make the visual claim that it does better on winnable drugs; the control arm is what shows that it does not.

aggregation, not representation: at fifteen cells the control-referenced shift cosine reaches 0.793 [0.775, 0.809]. The drug is not hiding in the magnitudes, which matters twice over. It rules out the simplest alternative explanation for everything that follows, and it means the repair of Section 4.6 cannot be read as “restoring the magnitudes”: that repair keeps rank throughout and changes only *what is being ranked*. The same measurement bounds the task from below — at single-cell resolution the same-drug versus different-drug gap is +0.002, so whatever a model can learn here, it cannot be learned from a single cell.

Third, *does the scramble swap in similar drugs?* The swap-distance sweep resolves this: Δ_{scr} is null in every distance bin including the far tail (−0.019), across a $2.3\times$ spread in swap distance, with a trend correlation of +0.051. Telling the model it is a drug whose true response is maximally different still does not change the output.

4.3 Where the blindness lives

A null result invites two readings, and they call for opposite responses: either the model cannot do the task, or the task cannot be done. The previous section closed the second — the ceiling rises with aggregation, a third of conditions are identifiable in principle, and the model is at chance even on those. So the failure is the model’s, and the useful question becomes *where* in the model it happens. That question cannot be answered behaviourally; it needs the weights.

Drug identity *is* in the representation: linear probes recover it at 82% (layer 9) and 76% (layer 16) against a chance rate of $\approx 8\%$. With cell line, plate and dose projected out, decodability *rises* with depth, from 0.354 at layer 2 to 0.883 at layer 16. Yet the purified drug subspace’s variance share stays flat at ≈ 0.03 , against ≈ 0.6 for cell line.

Layer	Decode (raw)	Decode (context removed)	Ablate pure drug (KL)	Random matched dim	Ratio
2	0.408	0.354	0.155	0.012	12.7×
4	0.585	0.429	1.563	0.084	18.6×
6	0.692	0.529	1.435	0.251	5.7×
8	0.779	0.569	0.596	0.042	14.1×
9	0.821	0.602	0.418	0.100	4.2×
12	0.754	0.873	0.133	0.035	3.8×
16	0.758	0.883	0.015	0.004	3.6×

Table 6: All seven layers pass the removal gate (88–95% of above-chance drug signal removed); chance is 0.083. Two decode series are shown because they behave differently and the distinction matters: the *raw* probe peaks at layer 9 and then declines, whereas with cell line, plate and dose projected out decodability rises monotonically to layer 16. The identity test is reported separately below, on a later and differently controlled design; the ablation ratios here are a statement about *functional variance* and carry no claim that drug identity is read.

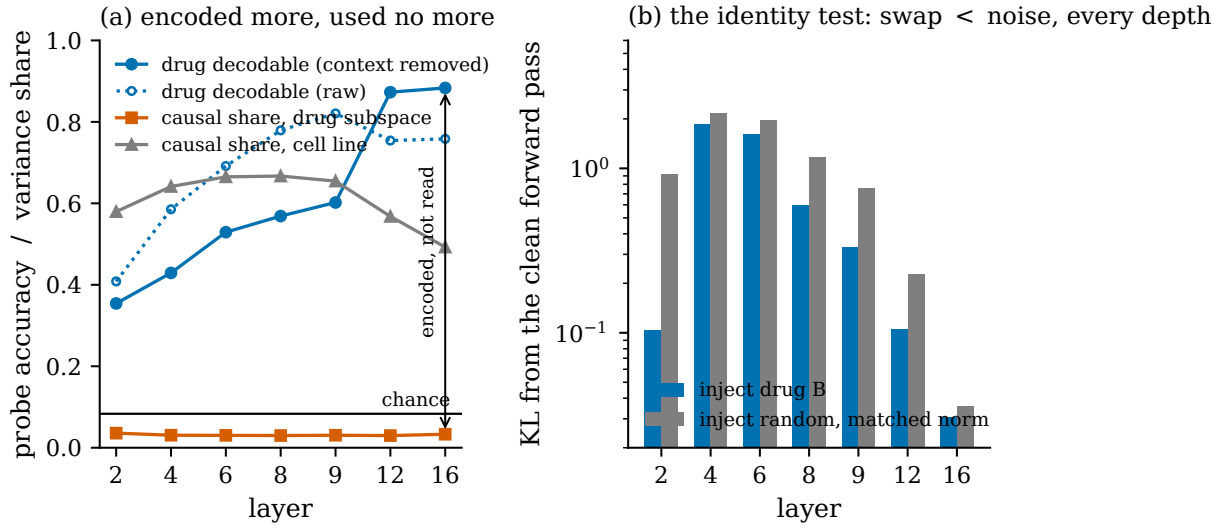


Figure 6: **The model computes an increasingly explicit drug representation and sets it aside.** (a) With cell line, plate and dose projected out, drug decodability rises monotonically with depth, from 0.354 at layer 2 to 0.883 at layer 16, against a chance rate of 0.083; the causal variance share of that same purified subspace stays flat near 0.03, against roughly 0.6 for cell line. The raw probe is drawn too and behaves differently — it peaks at layer 9 and declines — because the two series are easily conflated and the dissociation rests on the context-removed one. (b) Ablation of the purified drug subspace against a matched-*dimension* random subspace, on a log axis because late-layer divergences are small for everything. This is a functional-variance comparison: it establishes that the subspace is not inert, and it makes no claim about whether drug identity is read, which requires the substitution test reported in the text.

The identity test in this section has been rebuilt twice, and the failures are reported because they bear on how much the replacement can carry. The first compared a drug-B injection inside a ≤ 10 -dimensional purified subspace against a random vector drawn in the full 2048-dimensional stream, so roughly 99.5% of the control’s energy lay in directions the intervention never touched. The second drew the control inside the subspace, which looked like a repair and was not: the hook ablates before adding, so what is delivered is (added – removed), and an uncentred class mean shares its global-mean component with the thing removed. The substitution arm was then close to a restoration and the control arm a real displacement. Simulated with no model present, the control delivers $5.3\times$ the perturbation in 100% of draws, with the predicted gap $2\langle c_B, s \rangle = 9.179$ against an observed 9.162. Both retracted results said the substitution moves the output *less* than the control, and both were reporting geometry. What survives untouched is everything not resting on the substitution: the ablation ratios, whose null was always a matched-*dimension* random subspace and correctly constructed; the variance shares; and the decodability-versus-depth dissociation in Table 6.

The replacement design of Section 3.5 was run across five training arms and, for the two residual arms, replicated at three further seeds. Of 26 headline tests — five arms by up to seven depths — exactly one survives Benjamini–Hochberg correction: the residual arm at layer 12.

Arm / seed	effect	95% CI	<i>p</i>
residual, seed 42	+0.0197	[+0.0063, +0.0335]	0.0010
residual, seed 43	+0.0118	[+0.0060, +0.0173]	0.0003
residual, seed 45	+0.0085	[+0.0029, +0.0143]	0.0015
single-cell arm, six depths	nothing detectable at any depth (median -0.0012)		

Table 7: The identity test, as a fraction of that model’s own clean preference gap between the two responses being compared. Seed 44 produced no layer-12 measurement — the instrument was excluded there for lack of dynamic range — so the pre-registered criterion of two replications in three new seeds is met with the third a non-measurement rather than a failure.

The value of this is that it is a *positive control on real data*. The residual arm is independently known to use the drug: it is the arm whose prompt sensitivity is established behaviourally in Section 4.7. The probe detects identity reading there, replicably, and detects none anywhere in the arm that is behaviourally drug-blind. A null from an instrument that has never been shown to register a positive is uninterpretable; this one has.

The claim the evidence supports is that the generation readout consults the drug representation *at very low gain*: about 1% of the preference gap, an order of magnitude below the pre-registered margin for calling an effect material, in the one arm where it is

detectable at all. It is not that the readout cannot read drug identity, and it is not the stronger mechanistic story that earlier drafts of this thesis told. Three limits belong with it. The effect appears at one depth of seven, and layer 4 is positive in a single seed and null in the other three, which is what multiplicity discipline is for. The single-cell arm has no layer-12 measurement at all — it was excluded there for instrument saturation — so the cleanest comparison, the same depth across two arms, does not exist; the dissociation rests on that arm being null at all six depths it could measure. And the sibling residual-holdout arm does not reproduce the effect. Read together with the decodability series, the picture is a representation that becomes more explicit with depth while the process that writes the output barely consults it — which is suggestive of the workspace account of Section 2.6, and is offered as an interpretation of the pattern rather than as a demonstration of the mechanism.

4.4 The ε -ladder: no target-side fix works

If the defect is a readout that feels the magnitude of activity in the drug region but not its direction, then improving what sits *in* that region cannot help — and improving the training target is exactly that kind of intervention. The obvious objection is that one failed target construction proves nothing, so rather than try another we swept the family: entropic regularisation gives a one-parameter path whose two endpoints are the two constructions already tested independently, which turns a sequence of attempts into a controlled sweep.

Target	ε	Δ_{scr}	Verdict
single cell (real treated cell)	≈ 0	+0.014 [−0.016, +0.042]	null
OT barycentric (state-matched)	0.5	−0.001 [−0.018, +0.016]	null
consensus (global mean)	$\rightarrow \infty$	−0.010 [−0.022, +0.003]	null

Table 8: The regularisation parameter interpolates between the two target constructions tested independently, and no point on the ladder lifts the model out of the near-null regime. These figures come from a *random-partner* scramble; Section 4.6 shows that under a sharper stratified test the optimal-transport arm does show a weak effect (+0.0263 [+0.0069, +0.0476]), roughly five-fold below what re-encoding the target achieves. The consensus endpoint is additionally the ladder’s weakest-evidenced point: that arm was stopped at roughly 60% of an epoch and its tier-1 NIR was unscorable, so the negative rests on the single-cell and optimal-transport endpoints.

Training was not inert: the consensus model stops echoing the control (0.653 against a control-copy of 0.766, where the single-cell model sits *on* the control at 0.771) and its predictions are not mode-collapsed. But the variation is uncorrelated with real drug identity (Mantel $\approx$ +0.03). Not collapse — wrong structure. Denoising changes *what is in* the target while the defect is what its tokens encode; that is the quantity the next two

sections measure and then change.

4.5 How much signal is there to encode? The transfer coefficient

Before re-encoding anything, it is worth knowing how much drug-specific signal exists and what kind it is, because the answer scopes every claim in this thesis. The residual decomposes into a per-drug constant $\beta(d)$ and a drug $\times$ cell-line interaction $\kappa(d, c)$ (Section 3.10), and the two components call for different responses: the first is retrievable without any model, the second is the only thing a conditional model could win on. NIR cannot adjudicate between them, so the question is posed in cosine space with the measurement noise divided out.

The measurement itself carries a control-design lesson, stated because it changed the protocol. The first run compared $\mathcal{T}$ against a negative control of *different drugs on the same plate*, which returned +0.485 against $\mathcal{T} = 0.511$, and was correctly declared void: that control differs from the estimand in two ways at once — different drug *and* same cell line — and plate structure surviving a cell-line-scoped generic inflates same-plate comparisons (to +0.478) while leaving cross-line comparisons untouched. A *structure-matched* control — different drug, different cell line, breaking only the drug match — is the one that bounds the estimand, and a negative control that differs in more than one way cannot do so.

Quantity	Estimate	95% interval	Reference
$\mathcal{T}$ (cross cell lines, plate-scoped generic)	0.558	[0.514, 0.593]	simulated $\kappa=0$ null ≈ 1.0
drug $\times$ cell-line interaction share	44.2%	[40.7%, 48.6%]	—
$\mathcal{T}$ across dose (same drug, same line)	0.707	[0.632, 0.782]	reference scale
matched negative controls	−0.017, −0.008	spanning zero	must be ≈ 0

Table 9: The transfer coefficient, estimated on real molar doses with dyadic cluster-robust intervals (Section 3.10). The matched negative control sits on zero; the same-plate control that produced the earlier false alarm reads +0.478 for the reasons given in the text.

Roughly 44% of the drug-specific residual variance is drug $\times$ cell-line interaction, robust to the generic’s scope and to control splitting. Two internal checks agree, and both were decisive between scopes. *Dose ordering*: same-drug, same-line, different-dose transfer is 0.707 [0.632, 0.782], above the cross-line 0.558 — changing the dose costs *less* than changing the cell line, the biologically correct ordering; at cell-line scope the ordering inverts. *Control-split bias*: at plate scope the shared- and split-control builds agree to −0.000, against +0.254 at cell-line scope, so that discrepancy was plate contamination rather than the split. The reliability that enters the disattenuation is a within-well sampling precision (Section 3.7), optimistic relative to true biological reproducibility; the optimism sits in the denominator, so the interaction share is if anything understated. The per-drug

constant $\beta(d)$ — which the lookup of Section 4.8 reproduces almost exactly — is therefore *not* the whole signal. There is a conditional target worth about half the drug-specific variance that a per-drug constant cannot express.

Whether anything could *learn* that interaction is a separate question, and the direct test is sobering. Writing $\kappa(d, c) = r(d, c) - \hat{\beta}(d)$ with $\hat{\beta}$ estimated leave-one-cell-line-out, and disattenuating by κ ’s own split-half reliability:

Configuration	same cell line	different cell lines	excess within-line
plate-scoped generic	−0.004 [−0.011, +0.002]	+0.003	−0.007
cell-line-scoped	−0.003 [−0.021, +0.015]	+0.002	−0.005

Table 10: No cell line bends different drugs in a consistent direction. 4,000 pairs per arm; the signal interval at plate scope excludes anything above +0.002, so this is a real null rather than an underpowered one. The cell-line-scoped row is wider ([−0.021, +0.015]) and carries the claim less strongly.

We looked for structure in the interaction and did not find it: no cell line modulates different drugs in a consistent direction, and a direct test of whether mechanism-matched drugs share an interaction with a given cell line returned nothing detectable (permutation $p \approx 0.15$ – 0.31). The temptation is to read two negative results as *irreducibility*, and they do not support it. The second test bounds its own power severely: κ ’s split-half reliability in this atlas is 0.195, so the quantity being correlated is barely estimable, and same-mechanism compounds are so heavily co-plated that only 3–7% of matched pairs survive a different-plate requirement, which leaves the plate confound in place. The defensible reading is that κ ’s structure is *unresolved in this dataset*. Resolving it would take more cells per condition, or a plating design that crosses mechanism with plate — neither of which is an analysis choice.

The same measurement scopes the whole thesis. The shift’s energy decomposes exactly into residual 0.621, generic 0.390 and a cross term of −0.011: 62.1% of the perturbation response’s energy is drug-specific, 39.0% is the generic program, and the two components are essentially orthogonal, so the shares are genuine. Every claim in this thesis is scoped by the first of those numbers, and it has not previously been computed for this task. It also forces a retraction: the justification for estimating the generic per cell line rather than per plate was a reproducibility comparison, 62% against 19%, measured with *shared* control cells — which inflate cell-line reproducibility far more than plate reproducibility. Measured with split controls the two are comparable (20% plate, 16% cell line), so the argument favours neither, and the dose-ordering and control-split diagnostics above favour plate outright. The production targets are therefore built at plate scope (Section 3.7).

4.6 Changing what the tokens encode

Every target-side intervention changed *what the target was* and none changed *what its tokens encode*. That distinction is measurable rather than rhetorical: because a cell sentence ranks genes by expression, the target strings for two drugs in the same context are nearly identical, and the loss weights every token position equally.

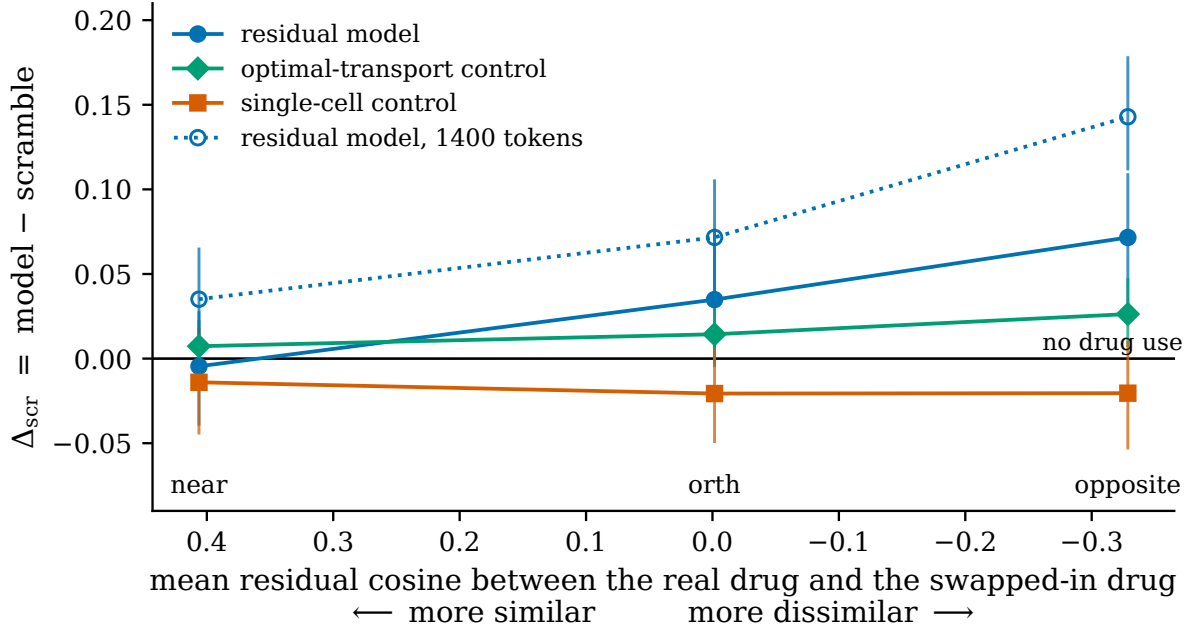
Representation	Inter-drug top-200 tokens differing	Replicate retrieval ceiling
full profile (the standard target)	34.6 / 200 (17%)	0.742
shift (treated – control)	111.4 (56%)	0.742
residual (drug-specific)	118.2 / 200 (59%)	0.7432

Table 11: 83% of the standard target’s tokens are identical between any two drugs in the same context. The residual row is quoted at **cell-line** scope; at plate scope it reads 120.3 and 0.7467. Note the retrieval ceiling is *unchanged* across all three rows: the information was always present, and the tokenisation discarded it.

The natural inference is that only a small fraction of the gradient can be carrying drug identity. That inference is a *hypothesis* about the optimisation: what is measured here is the overlap of the target gene sets, not a token-level loss or gradient attribution, and it should be read at that strength. Its warrant is predictive — it explains why every change to target *content* failed identically, and it predicts that changing what the tokens *encode* would not. Re-encoding the target as the drug-specific residual raises the differing fraction to 59% at an unchanged retrieval ceiling, and produces the first non-null result in the project. On the earlier target build, the stratified scramble gave a gap that grew with swap dissimilarity — +0.0351 [+0.0061, +0.0656], +0.0716 [+0.0371, +0.1059] and +0.1429 [+0.1112, +0.1787] across the **near**, **orth** and **opposite** strata — while the single-cell and optimal-transport controls stayed flat and null at every stratum (−0.001 to +0.016). The monotone gradient is itself the evidence: a metric artefact would not order itself by swap dissimilarity.

Three further diagnostics on that build agreed, and only the residual model passed all three: the correlation between condition reproducibility and model NIR was +0.144 (the model does better where the drug effect is real) against −0.117 and −0.216 for the controls; prediction diversity was highest for the residual model; and while $\cos(\text{prediction}, \text{own truth})$ exceeded $\cos(\text{prediction}, \text{other drugs})$ for all three arms, the residual model’s margin (+0.084) was roughly twice the optimal-transport arm’s (+0.049) and three times the single-cell arm’s (+0.026).

Two measurement lessons belong with this result. The effect was first measured at a 600-token generation cap, which silently truncated 26% of generations — a signature is ≈ 201 gene symbols at 2–4 BPE tokens each — and *halved* it (+0.0716 against the correct +0.1429); re-running both controls at the matched 1400-token budget left them flat and null, so the comparison is budget-matched and the conclusion unaffected. And the



residual frame. Intervals are 95% bootstrap, clustered over cell lines; $n = 250$ conditions per point.

Figure 7: **On the earlier target build, the residual model’s advantage over its own scramble grew with how dissimilar the swapped-in drug’s response was; a control scored through the identical pipeline stayed flat and null.** The evidence is the gradient, not any single point — a metric artefact would not order itself by swap dissimilarity. The horizontal axis is the measured mean residual cosine between the real and swapped-in drug, which is byte-identical across all four runs, so the positions are exact. The three solid series are at a matched 600-token generation budget; the residual model’s 1400-token run is drawn separately as the dotted overlay, since the budget alone doubles the effect. Intervals are 95% bootstrap clustered over cell lines, $n = 250$ conditions per point. The **near** stratum is the weakest test by construction: where the swapped-in drug’s true response is similar, an unchanged output is correct rather than blind. These magnitudes are superseded by the rebuilt-target run of Section 4.7; the figure stands as the first demonstration of the effect.

optimal-transport arm, originally reported as achieving nothing, scores $+0.0263$ $[+0.0069, +0.0476]$ under the sharper stratified test with a monotone gradient: it does use the drug, weakly. The original null was an underpowered instrument, not an absence of signal; the ε -ladder conclusion stands, but “OT achieved nothing” is withdrawn.

What those magnitudes can and cannot claim is re-adjudicated in the next section, because they were read against the **opposite** comparator, and that comparator turns out not to be a null. The comparator-free evidence is already in hand, and it is the strongest in this thesis: regressing each generation’s NIR on the cosine between the named drug’s truth and the target’s truth gives a slope of $+0.390$ $[+0.225, +0.542]$ on trained conditions — zero slope is the natural null, so no comparator defect can reach it. Re-encoding the target made the model respond to the combined drug-and-mechanism prompt. What that response is worth is the question the rest of the chapter answers.

4.7 Sensitivity is not fidelity: a main effect, not an interaction

A model that responds to the drug token has cleared a low bar, because reproducing a training average requires responding to it too. Everything scored above came from conditions the model trained on, so the result so far shows that the token is consulted — not that anything transferable was learned. Separating those two requires withholding data at the right granularity (Section 3.8), and it requires a comparator that is genuinely uninformative. The first finding of this section is that the comparator used up to here was not, and every magnitude quoted against it must be re-read.

The comparator’s own score gives it away. Across the three swap strata, the scrambled model’s NIR is 0.572, 0.497 and 0.403 as the named partner’s true response moves from aligned ($\cos = +0.26$) through orthogonal (0.00) to anti-aligned (-0.24) with the target. A comparator whose score is a monotone function of which drug it was handed is not a null: it is an active treatment that tracks the partner drug. The swap partner is itself a drug the model was mostly trained on, so telling the model a lie costs it precisely because it knows the partner. A gap measured against such an arm decomposes as

$$\Delta_{\text{scr}} = \underbrace{(\text{NIR}_{\text{model}} - 0.5)}_{\text{how much naming the truth helps}} + \underbrace{(0.5 - \text{NIR}_{\text{scramble}})}_{\text{how much naming the lie hurts}}, \quad (16)$$

and only the first term is wanted. Against **opposite** the second term is $+0.097$; against **orth**, whose own score sits on chance (0.497), it is zero. Only the neutral column of Table 12 can therefore carry a magnitude claim, and the **opposite** column is reported solely as the diagnostic that would have misled us.

The control behaves exactly as the diagnosis predicts. On **unseen_drug** — drugs held out of fine-tuning, where no learned knowledge can be at work — the gap against **opposite** is $+0.0862$ $[-0.0028, +0.1753]$, nine times the neutral estimate, with an interval

Split	n (cell lines)	model NIR	Δ_{scr} vs orth (neutral)	Δ_{scr} vs opposit
train	200 (35)	0.635 [0.575, 0.695]	+0.1457 [+0.0836, +0.2078]	+0.2627 [+0.21, +0.31]
unseen_combo	250 (39)	0.545 [0.493, 0.596]	+0.0449 [−0.0074, +0.0972]	+0.1281 [+0.07, +0.19]
unseen_drug	120 (38)	0.512 [0.442, 0.581]	+0.0097 [−0.0655, +0.0848]	+0.0862 [−0.00, +0.17]

Table 12: Whole-well holdout on the rebuilt targets (split-before-fit, plate-scoped generic). Only the **orth** column can carry a magnitude claim; the **opposite** column adds “how much naming a lie hurts” to “how much naming the truth helps” and is shown as the diagnostic. The **unseen_drug** row is the control that proves the point: those are drugs the model has never seen, its gap must be zero, and it is zero only in the neutral column. Model NIR intervals test against the chance value 0.5 with no comparator. Intervals are two-way cluster-robust over cell lines and wells, with a Student t reference (Section 3.4).

that only just contains zero. The same conditions, the same generations and the same model scored against the neutral stratum give +0.0097 [−0.0655, +0.0848], sitting on zero as a control should. The borderline interval on the **opposite** arm is itself a lesson in the inference regime: a cluster-robust variance is asymptotic in the number of *clusters* rather than observations, this arm spans only 25 treatment wells, and under the correct t_{24} reference (2.06 rather than 1.96) the interval widens by five per cent and covers zero. An earlier draft of this section read the normal-quantile interval as excluding zero; it does not.

Read in the neutral column, **train** clears zero at +0.1457 [+0.0836, +0.2078] and **unseen_combo** does not: +0.0449 [−0.0074, +0.0972], spanning zero under well-clustering and line-clustering alike ([−0.0050, +0.0993]). Generalisation to an unseen treatment well of a seen drug is therefore *not established*. The precision matters: not established is not the same as shown to be zero. The interval reaches +0.097, so an effect of up to roughly +0.1 remains compatible with these data at $n = 250$ — a limit of power at the sample size the holdout affords, reported as one. (On **unseen_drug** the corresponding detection limit is +0.099.) The difficulty of the three splits is not the explanation: their split-half replicate ceilings are 0.962, 0.955 and 0.956, so the splits are equally hard and the held-out shortfall is not a noise-floor difference.

The difference between the trained and held-out estimates is +0.1008 [+0.0196, +0.1820] with cluster-robust $p = 0.0150$ (an unclustered permutation returns $p = 0.0054$, quoted only for contrast, because ignoring the clustering this thesis mandates everywhere else flatters the result). advantage *is* requires care, because the two arms differ in two ways at once: whether the condition was trained on, and *which drugs they contain*. Restricting the comparison to the 20 drugs present in both arms, the advantage attenuates to +0.0420 with $p = 0.4764$. The defensible statement is therefore narrow: a condition-weighted training-exposure advantage exists in this evaluation, and its generality across drugs is unresolved — the design cannot separate exposure from drug identity, and we do not claim either direction of the stronger interpretation.

Two instruments that need no comparator at all cannot be touched by any of this, and they carry the positive content of the section. First, the slope test: regressing each generation’s NIR on the cosine between the named drug’s truth and the target’s truth gives +0.390 [+0.225, +0.542] on **train**, +0.374 [+0.214, +0.508] on **unseen_combo** and +0.238 [+0.033, +0.420] on **unseen_drug** — all three exclude zero. The model emits something aligned with the named drug’s signature on every split, including held-out wells, with no degradation worth the name between train and held-out. Second, scored against chance with no comparator, the model’s own NIR clears 0.5 on **train** (0.635 [0.575, 0.695]) and is not distinguishable from it on either held-out split. A positive slope alongside a chance-level NIR is not a contradiction: it says the drug-to-signature map exists but is *coarse*. The model knows roughly what a drug does and not what it does *in this well* — and NIR asks the second question, ranking against the ≈ 40 other drugs measured in the same cell line. That is the drug main effect without the drug $\times$ cell-line interaction, which Section 4.5 measures at 44.2% of the residual variance. It is also why the lookup wins in the next section: **drug_lookup** *is* that main effect, estimated straight from data far more precisely than the model learned it.

The directional-agreement measurement says the same thing at condition level. The cosine between the model’s prediction and its own condition’s truth on trained conditions averages +0.0511 (sd 0.1141, median +0.0300), with individual conditions spanning [−0.1819, +0.4395] and 32.0% clearing |0.1|: weak on average and substantially heterogeneous, so a single summary number would misrepresent it in either direction.

One caveat attaches to every held-out number in this chapter, and it is stated here where those numbers live. The evaluation’s truths are reconstructed through a compatibility path that refits the generic on the reliability-surviving subset of training conditions rather than the full set the checkpoint was trained against, and that filters held-out conditions by their own split-half reproducibility — a selection on the evaluation set’s outcome. Its direction is conservative for the headline: it retains the cleanest, easiest conditions, and the held-out result still fails to clear zero. But the evaluated population is not the full held-out population, and the intervals above should be read as applying to the reproducible subset.

Two instrument defects, both of which would have concealed everything in this section, are worth recording because they are now checks rather than assumptions. A uniform condition sampler reproduces the pool’s split proportions, so 250 constructed held-out combinations yielded only 33 scored ones; and a strict-inequality NIR scored the zero vector (“predict the generic response”) at 0.000 rather than 0.500. Both are corrected (Section 3), and after correction the drug-agnostic baselines land where they must: **generic** exactly 0.500, **control_copy** 0.511, **random** 0.521. The residual frame is *demonstrably* leak-proof rather than assumed to be.

The sentence the evidence supports, then, is narrower than the one this section origi-

nally carried: re-encoding the target made the model respond to the combined drug-and-mechanism prompt and learn a drug-level average response. It learned the main effect and not the interaction, and its generalisation to unseen wells is a power-limited open question rather than a result in either direction.

4.8 But a lookup table wins

Transfer without fidelity is a real property, but it says nothing about quality: a model can apply a learned signature in a new context and still apply it badly. The pre-registered bar was therefore never “beats chance” — residual targets are plausibly memorisable, and a predictor that simply looks up what this drug did elsewhere needs no model at all. That predictor is the comparison that matters.

Arm	NIR	note
ceiling (within-condition split-half replicate)	0.958	0.962 / 0.955 / 0.956 by split
<code>drug_lookup</code> (training-only)	0.941	the bar to beat
<code>drug_lookup_1</code> (one other line)	0.859	removes the averaging advantage
model	0.569	0.635 / 0.545 / 0.512 by split
random	0.521	floor
<code>control_copy</code>	0.511	drug-agnostic
<code>generic</code>	0.500	the zero vector, by construction

Table 13: The baseline ladder on the rebuilt targets, $n = 570$ scored conditions; the lookups are fitted from training conditions only (Section 3.9). The paired comparison on the common support ($n = 449$) is model – `drug_lookup` = -0.3559 [-0.3988,-0.3130], and of the distance from chance to the replicate ceiling the model covers 18.6% against the lookup’s 96.3%.

The pre-registered interpretation, written before the result, anticipated three branches; the middle one fired. The model reads and transfers the drug, but recovers only a small fraction of its signature. On the earlier build it also beat `moa_lookup` (0.529, $n = 143$), so its prompt response is not reducible to the **Mechanism:** field — though at that level the bar is low.

This is DrEval’s finding, in transcriptomic space. DrEval reports that deep learning models barely outperform a naive predictor of mean drug and cell-line effects [13]. Their **NaiveMeanEffectsPredictor** is $\hat{y}_{ij} = \mu_i^c + \mu_j^d - \mu$; our `drug_lookup` is the $\beta(d)$ term of that same additive model, measured in a frame where the generic(c) term has already been subtracted. The finding replicates and sharpens: on a 946-gene transcriptomic response rather than a scalar potency, and against a proper within-condition replicate ceiling, the additive drug main effect reaches 0.941 of a 0.958 ceiling while a 1B-parameter model reaches 0.569. Note also that the warning DrEval inherits from [18] — that good pair-holdout performance is reachable by predicting each drug’s training average — is treated

here as the hypothesis rather than the caveat, and measured. It holds.

It is tempting to read $\text{ceiling} - \text{drug_lookup} = +0.017$ as “the modelling target has vanished”. That inference is not supported, for three reasons that all favour the lookup: the ceiling is scored against a *half-sample* truth while every baseline is scored against the *full-sample* truth; drug_lookup averages ≈ 38 conditions against the ceiling’s single noisy half; and NIR near 0.96 is compressive, collapsing a wide range of cosines into a few thousandths. The paired gap on common support, -0.3559 $[-0.3988, -0.3130]$, is the honest summary of the contest; the coverage figures, 18.6% against 96.3% of the chance-to-ceiling distance on that same support, are its scale-free restatement.

4.9 Every arm in one place

The results so far are distributed across two measurement frames, and a reader assembling them has to keep track of which numbers may be compared with which. Table 14 collects them.

The two frames are *not* interchangeable and are separated deliberately. The expression frame scores a predicted cell profile against other drugs on the same plate; the residual frame scores a predicted drug-specific signature against other drugs in the same cell line, after the generic program has been subtracted. Both have chance at 0.50 by construction, and that coincidence is the trap: the residual frame’s ceiling is 0.958 where the expression frame’s is 0.576, so a model scoring 0.569 in one and 0.498 in the other has not improved — it is being measured against a different question.

Frame	Arm	NIR	What it knows
expression (within plate, tier 2, $n = 606$)	noise ceiling	0.576	a real replicate of this condition
	control-copy	0.504	nothing — it reproduces the untreated cell
	linear map	0.500	the control cell, no drug
	model	0.498	control cell + drug
	global mean	0.180	nothing
residual (cell-line sets, $n = 570$)	noise ceiling	0.958	a real split-half replicate of this condition
	drug lookup (training-only)	0.941	this drug’s signature from other cell lines
	drug lookup, one line	0.859	the same, from a single other cell line
	model	0.569	control cell + drug
	scramble, orth partner	0.497	the neutral comparator: chance by construction
	scramble, opposite partner	0.403	an active treatment, not a null
	generic response	0.500	nothing (the zero vector, by construction)

Table 14: Every arm on the calibrated metric, by frame. Chance is 0.50 in both. The residual-frame figures are the rebuilt-target run of Sections 4.7 and 4.8; the held-out subset carries the outcome-selection caveat stated there.

Read down the residual block and the thesis’s shape is visible in one object. A replicate of the experiment reaches 0.958; a lookup table reaches 0.941; the model reaches 0.569;

the same model told a response-orthogonal drug sits at 0.497, the neutral zero point, and told an anti-aligned drug falls to 0.403. The distance between the model and its neutral scramble is the drug effect this work recovered; the distance between the model and the lookup is what it did not.

4.10 Scope: what is closed by measurement

An unseen drug can only be predicted through a property it *shares* with drugs seen in training, so the question is answerable by enumerating the channels and measuring each in the same frame. For each channel we predict a drug’s residual from the mean residual of the *other* drugs in the same cell line that share the property — the construction of `moa_lookup`, generalised — and pair it with a plate-matched random null, because the estimand requires the two arms to differ in exactly one respect.

Channel	Coverage	<i>n</i>	vs plate-matched null	(vs count-matched)	Verdict
protein target	54%	894	+0.0973 [+0.0741, +0.1183]	+0.1269	live
mechanism (MoA)	32%	1053	+0.0749 [+0.0538, +0.0948]	+0.1067	live
chemical structure	100%	4084	+0.0162 [+0.0051, +0.0264]	+0.0206	closed

Table 15: Coverage is stated first: a channel annotated on few drugs cannot be *closed* by a null result, because it was never tested. The plate-matched null is the estimand-correct control; the count-matched column shows how much of each raw gap was co-plating. Chemistry reproduces the structure gate in this frame; both knowledge channels clear their corrected nulls.

The size of that correction is itself informative, so it is reported rather than argued away. Mechanism partners are co-plated with the target 0.420 of the time against 0.340 for a random draw (protein target 0.348 against 0.323; chemical similarity 0.344 against 0.345 — none, as expected for a channel uncorrelated with plating). Under a third construction that draws all partners from *other* plates, the knowledge channels’ advantages fall to +0.0643 and +0.0344 and chemistry’s interval spans zero (+0.0056); and in that construction the channel arms’ absolute NIR sits at 0.360 and 0.296, *below* chance — a direct demonstration of how strongly the cell-line-scoped residual frame is dominated by plate. The verdicts above survive the correction; the magnitudes shrank by a fifth to a third, which is what the co-plating fractions predict.

This overturns a pre-registration made in this project. Two design documents concluded that unseen-drug generalisation was out of reach, and both inferred it from the structure gate — that is, from Morgan fingerprints and a molecular language model. But the drug-knowledge specification written earlier in the same project ranks raw structure the *lowest*-value channel and protein targets “the key feature”, on the reasoning that transcriptional response is driven by which protein a compound hits, something structure encodes only indirectly. The pre-registration generalised from the weakest channel to the

strongest, and it was wrong. Tahoe’s own metadata carries protein-target annotations for 264 drugs — 280 distinct symbols, 550 edges — which no analysis in this project had read.

The correct statement is therefore narrower and more useful than the one we started with: *chemical structure does not predict response, measured twice in two frames; protein target and mechanism do*. Unseen-drug generalisation is reachable through drug-side *knowledge*, and not through chemistry. Three caveats bound it. Coverage is 54% and 32%, so this is measured on the annotated subset. The channel gaps are severalfold below the advantage a same-drug lookup enjoys for a *seen* drug (+0.44 above chance), so the reachable ceiling in this regime is much lower. And target and MoA overlap, so whether target adds anything *over* MoA is a further question this measurement does not settle — settling it would need a leave-one-mechanism-out split rather than the leave-one-drug-out split used here, since a held-out drug’s mechanism class is generally still represented in training.

4.11 The channel the model already has

The natural next step from the preceding section writes itself: append the protein targets to the prompt, retrain, and see whether unseen-drug performance rises. That arm assumes the bottleneck is *information availability*. Everything in this thesis points the other way, but the direct evidence is weaker than earlier drafts claimed, and this section exists to say exactly how weak it is.

The scramble has until now replaced the drug name *and* the mechanism together, so a non-null gap establishes sensitivity to the combined prompt without attributing it to either span. On the earlier target build, a decomposed version suggested the response is carried almost entirely by the name span: swapping only the drug name gave $\Delta_{\text{scr}} = +0.0809 [+0.0592, +0.1010]$ ($n = 570$), swapping only the mechanism gave $+0.0091 [-0.0174, +0.0383]$ ($n = 325$), and both together $+0.1024 [+0.0827, +0.1236]$. Those arms were scored against the **opposite**-signature comparator that Section 4.7 shows inflates every gap read against it, and they were not re-run under the corrected evaluation. The claim that the model “reads the name and not the mechanism” is therefore withdrawn pending that re-run; what the current run establishes is response to the combined drug-and-mechanism prompt, with the mechanism field’s contribution unresolved.

The practical consequence is narrower than the one this section originally drew, and more honest. The target-prompt arm is not retired by measurement — it is untested. What argues for building it carefully rather than eagerly is the shape of the failure documented above: the model’s deficit is not confined to missing information about unseen drugs, because on *seen* drugs it recovers only a fifth of the chance-to-ceiling distance against a lookup that knows nothing about contexts at all. And what argues for building

it at all is the one thing a lookup structurally cannot do: handle a drug it has never seen, in the one regime where Section 4.10 shows live channels exist. The model’s opportunity is not the interaction — Section 4.5 found 44% of the residual variance there but no learnable structure in it — it is the unseen drug. Whether the model can be made to take that opportunity is the question the thesis leaves open.

5 Limitations and Future Research Directions

5.1 Limitations of scope

Drug coverage. The residual analysis is built on a cache of 620,564 cells covering 106 drugs across 50 cell lines, sampled from Tahoe’s shards — roughly a tenth of the atlas’s compound set. This is ample to test whether the model uses the drug, and to measure transfer, but it is not a population estimate. Any claim about the *distribution* of drug behaviours should be read against the broader characterisation, which used 6,628 conditions.

Only reproducible conditions. The residual arm retains the 62% of conditions whose residual reproduces across a half-split ($\cos > 0.2$). This is the right choice — training on irreproducible residuals is training on noise — but it means results describe the winnable fraction of the task, not all of it. The filter might be expected to select *toward* context-independent conditions and so bias the transfer coefficient upward. Measured, it does the opposite: the unfiltered estimate is 0.598 against the filtered 0.557. The filtered value is the one quoted, because disattenuation is unstable at the very low reliabilities the filter removes.

The intervals omit generation variance. Every confidence interval in this thesis is a clustered bootstrap that resamples *cell lines* while treating each condition’s score as fixed. It therefore captures between-cell-line variance and not the variance introduced by sampling the model’s output. With four generations per arm at temperature 0.8 that second source is not negligible, and it is not measured here. An across-seed replication — three seeds, identical in every other respect — would supply the missing component, and the honest form of every interval below would then be a point estimate with a seed spread beside it. **[TODO: run the three-seed replication and report the across-seed spread beside the within-seed interval, stating which sources each captures]**

A swing that looked like generation variance and was not. Two evaluations of the same checkpoint on the same conditions, differing only in that one carried two additional scramble arms and so consumed the sampling stream differently, returned Δ_{scr} on the held-out-drug split as +0.0195 [−0.045, +0.086] and +0.1114 [+0.051, +0.173]. This was originally recorded as evidence of seed instability, and that diagnosis was wrong — or rather, it named the trigger and missed the cause. Both figures are gaps against the *opposite* swap, which Section 4.7 shows is not a null: it contains a term measuring how far the *partner* drug was pushed the wrong way, a quantity that has nothing to do with the target and that changes with which partner happens to be drawn. An arm whose value is partly a nuisance quantity will move when the draw moves. Against the neutral

comparator the same control sits at $+0.0097$ $[-0.0655, +0.0848]$. Two mechanisms are involved and both are now addressed: generation was genuinely unseeded — `do_sample` advanced from global torch state, so the two arms of a contrast were drawn from different points of one stream — and each call now seeds its own generator from (condition, arm, batch); and the comparator has been corrected. The general caution above stands regardless; this particular swing is no longer evidence for it.

Pseudobulk resolution. Residuals are condition-level pseudobulks over ≥ 40 treated cells. The single-cell heterogeneity of the response is not modelled in this frame, and the project’s own earlier results show why: at single-cell resolution the drug moves a cell less than noise does ($\text{SNR} \approx 0.75$, median 0 differentially expressed genes at $\text{FDR } q < 0.05$).

Cross-plate absolutes. The residual-space scores are computed with comparison sets grouped by cell line rather than by plate. The *difference* Δ_{scr} is leak-immune, because both arms share the same control cell, but the absolute NIR values in that frame are not directly comparable to the within-plate tables elsewhere. **[TODO: rescore the generation-free arms within plate — this costs no GPU time]**

Frame coverage. The drug-specific residual is $\|r\|/\|s\| = 0.786$ of the response, against 0.620 for the generic program and batch, and the two are essentially orthogonal ($0.786^2 + 0.620^2 = 1.002$). So this frame covers 62% of the response *variance* and every result here should be read within that bound — a real but not total share. The remaining 38% is precisely the component the field’s absolute-accuracy metrics are dominated by, which is why a drug-agnostic predictor scores so well on them.

5.2 Limitations of the unseen-drug evaluation

Two defects, both of which must be stated rather than papered over. A third — the comparator — was found and corrected rather than merely stated, and Section 4.7 reports the arm against the neutral stratum throughout; the two below survive that correction.

It is not a clean held-out-drug design. The prompt contains `Mechanism: {moa}`, so holding out a drug still hands the model the held-out drug’s class label. The correct design is *leave-one-mechanism-out*. The leak is probably small — `moa_lookup` scores 0.529, so the mechanism label carries little about the response — but the design is wrong, and the arm’s proper use is as an instrument control rather than as the basis for a positive unseen-drug claim.

It is underpowered by construction. The confidence interval half-width on the `unseen_drug` gap is $\approx \pm 0.075$, against a maximum effect available through the mech-

anism channel of ≈ 0.033 . The arm therefore *cannot* detect the largest effect it could plausibly be measuring. This cuts both ways and the direction matters. As a *control* it is doing the job asked of it: the question there is whether the point estimate sits at zero when the model has no drug knowledge, and $+0.0097$ does. As *evidence of absence* it carries almost nothing, because an effect of the size the mechanism channel could deliver would sit comfortably inside the interval. The distinction is what licenses using the arm to validate the comparator in Section 4.7 while declining to use it to argue that unseen-drug transfer is zero.

5.3 Directions

The arm this work closed rather than opened. The obvious continuation of Section 4.10 is a channel-conditioned prompt: append protein targets to the instruction, retrain, and evaluate on a leave-one-mechanism-out split. We do not recommend it, and the reason is measured rather than argued. Section 4.11 shows the model reads the drug *name* ($+0.081$ [$+0.059, +0.101$]) and does not measurably read the *mechanism* field it is already given ($+0.009$ [$-0.017, +0.038$]), in an arm powered to detect an effect the size of the name’s. Adding a target field to a prompt whose mechanism field is ignored addresses information availability, and the binding constraint is the readout.

What would be worth building instead is an intervention that makes the drug *causal* rather than merely present — a learned per-drug modulation of the transformation from control to treated, so that the drug parameterises the computation instead of sitting in the residual stream as an input the readout can round away. That is a research architecture rather than a prompt change, and it is outside the scope of what remained here.

Two channel questions that do remain open. Target and mechanism overlap, so whether target adds anything *over* mechanism is unresolved — and it matters, because the prompt already exposes mechanism and the model already ignores it. Separately, the channel gate measures *similarity transfer*: an unseen drug resembling a seen drug with the same target. It does not test a *compositional rule* — that a named target’s own pathway moves, which would apply to any drug whose target is named, including one with no similar drug in training. The second is the stronger claim, it does not reduce to a lookup, and it is untested.

Unseen cell lines. A cell line never seen with any drug is a different generalisation axis from the held-out pairings tested here, and one not blocked by the structure gate: it asks whether the control cell’s state is used *compositionally* with the drug. The transfer coefficient now scopes it: with the interaction both substantial ($\approx 45\%$) and idiosyncratic, a new cell line offers no consistent direction to exploit, so the achievable target there

reduces to recalling the drug’s per-drug constant — which a lookup already does at 0.963. The axis is therefore less promising than it appeared before Section 4.5 was measured.

Objectives that were considered and rejected. Recorded so they are not re-litigated:

- **Reinforcement learning with a contrastive reward.** Its mechanism — a per-sequence reward, so the token-dilution ratio never enters the gradient — attacks a bottleneck that the residual encoding has already lifted ($34.6 \rightarrow 118.2$ differing tokens); and its reward optimises similarity to the true residual, which is precisely what `drug_lookup` already achieves at 0.963. Its best case is convergence to the lookup.
- **A differentiable profile-level contrastive loss.** The natural construction, $\text{score}(g) = \sum_i P(\text{token}_i = g) w(i)$, presumes $\text{gene} \approx \text{token}$, but 86.9% of the 946 panel genes share a first sub-token under this tokeniser (mean 3.254 BPE tokens per gene, only 8 single-token). Under teacher forcing the difficulty compounds: the residual target *is* the drug signature, so the true prefix identifies the drug within a handful of symbols and the gradient vanishes exactly at the drug-blind solution the loss exists to penalise.
- **Loss re-weighting toward differentially expressed genes.** Re-weighted cross-entropy against a single target contains no negatives, so nothing in it forces the output for one drug to differ from another’s. Given that the readout is direction-blind (Section 4.3), it addresses token dilution but not the actual defect.

One model size, and the smallest one. Every result here comes from a 1B-parameter backbone. C2S-Scale releases checkpoints up to Gemma-2 at 27B, and we tested none of them, so the most predictable question an examiner can ask — does it work if the model is bigger — has no answer in this thesis.

Two things can be said about what we would expect, and they point in different directions. The bottleneck this work identifies is not capacity. It is that roughly 1% of the training gradient reaches drug identity, because the target tokens are overwhelmingly shared across drugs (Section 4.6); a larger model receives the same diluted signal from the same objective, and the fact that re-encoding the target lifted the effect while leaving capacity untouched is evidence that the constraint binds where we say it does.

The genuinely different argument for scale is not capacity but *prior knowledge*. A 27B model pretrained on far more text plausibly knows, as a fact about the world, that a named compound inhibits a named kinase — knowledge a 1B model may simply lack. That matters here because Section 4.11 finds the model reads the drug *name* but not the *mechanism* field it is already handed, and a model with real pharmacological knowledge

is exactly the case where that might change. This is a sharper hypothesis than “bigger is better” and it is testable cheaply with a low-rank adaptation of a mid-sized checkpoint rather than a full fine-tune. We did not run it, and it is the first thing we would run with more time.

The starting checkpoint was never prepared for this task. We fine-tune from C2S-Scale-Pythia-1b-pt, which is pretrained on atlas data alone — CELLxGENE and the Human Cell Atlas, no perturbation corpus of any kind — and whose released capabilities do not include perturbation prediction. This cuts two ways and both should be stated. It is why the held-out splits are clean: a drug withheld here was never seen at any stage, so nothing in the pipeline can have memorised it. But it also means the failure we report is a failure of *this* starting point, and a reader may reasonably ask whether a checkpoint already tuned for perturbation prediction would behave differently. We cannot answer that from the public artifacts. The perturbation-specialised model in the same line of work is a Gemma-3 4B checkpoint that was not released, and the available C2S-Scale checkpoints are the pretrained ones. What we can say is narrower and worth separating from the broader claim: every arm in this thesis starts from the same base and differs only in the variable under test, so the *comparisons* between arms are unaffected by this choice even where the absolute level is not.

Generality. Everything here is one backbone on one atlas in one representation. Whether the readout-specificity failure is a property of cell-sentence models specifically, or of autoregressive conditional generation over transcriptomes generally, is open. The mechanistic protocol of Section 3.5 — purified subspace, removal gate, positive control, matched-norm swap — transfers unchanged to any model with an accessible residual stream, and is probably the most reusable artifact of this work.

6 Conclusions

This thesis asked whether a large language model fine-tuned on the largest available single-cell drug-perturbation atlas learns drug-specific biology. It does not — and the interesting content is in the chain of results establishing why, how far the failure can be repaired, and what bounds the repair.

The evaluation had to be fixed first. The metric the field standardises on is exploitable: a zero-information baseline placing every gene at the middle rank scores $DE-\Delta r = 1.000$, above both the noise ceiling and the model. Under a Dynamic Range Fraction analysis computed within plate, only NIR — a discrimination metric, in which the generic response cancels — is calibrated. Of the four prediction metrics tested, three are clearly inverted and the fourth is approximately neutral. This is the most portable result here, since it applies to any work scored on $DE-\Delta r$ -family metrics.

This also settles how the thesis enters the dispute of Section 2.3. Miller et al. [14] argue that reports of deep models failing to beat uninformative baselines reflect miscalibrated evaluation rather than model failure, and on genetic perturbation they support that with calibrated-metric benchmarks. We port their framework, confirm its methodological claim emphatically — metric choice dominates, and only the rank-based metric survives — and then find that on drug perturbation the model is at chance *on the metric their own analysis endorses*. That is the strongest form the sceptical result can take: it cannot be answered with the objection their paper raises. Their conclusion is not overturned so much as bounded — it does not extend from genetic perturbation to this task, and the per-drug-constant finding above suggests why.

The model is drug-blind, and the representation is not the reason. Four independent instruments agree that swapping the drug token changes the output by nothing. Yet drug identity is linearly decodable from the residual stream, rising to 0.88 by layer 16 with context removed, while the purified drug subspace’s causal variance share stays flat near 3%. The model computes an increasingly explicit drug representation and does not act on it — a dissociation between what is encoded and what is used.

How far that can be pushed toward a mechanism is limited, and the limit should be stated plainly. Two earlier attempts to localise the failure by activation swapping were retracted after their matched controls were found confounded, and what replaces them (Section 4.3) is a small, replicated, single-layer effect present in one training arm and absent in another. That supports the modest reading — the generation readout consults the drug representation at very low gain — and does not support a claim about privileged subspaces or global workspaces, which is offered here as an interpretive frame rather than a result.

The bottleneck appears to be the tokenisation, and it is repairable. Because a cell sentence ranks genes by expression, the targets for two different drugs in the same context are nearly the same string: 82.7% of the top-200 genes are shared, so only about a sixth of the target distinguishes one drug from another. Since the training loss weights every token position equally, the natural inference is that very little of the gradient can be carrying drug identity — but that inference is a *hypothesis* about the optimisation, not a measurement of it. What is measured here is the overlap of the target gene sets; no token-level loss or gradient attribution was computed, and the claim should be read at that strength. What follows is that the hypothesis makes correct predictions, which is the evidence for it. It explains a uniform pattern of failure: an entire ε -ladder of target constructions, from a single state-matched treated cell to a full pseudobulk collapse, stays in the near-null regime, because all of them change *what the target is* and none change *what the tokens encode*. (One arm, the optimal-transport target, does show a weak effect once a sharper stratified test is applied — $+0.0263$ [$+0.0069$, $+0.0476$] — but roughly five-fold below what re-encoding achieves, which is the distinction that matters.) Re-encoding the target as the drug-specific residual raises the differing fraction to 59% and produces the first non-null drug effect in this project, with a monotone gradient across swap dissimilarity and a matched control that stays null.

What that drug use amounts to: a main effect, not an interaction. An earlier form of this paragraph claimed the drug use generalises, on a holdout keyed at the (drug, cell line) level where pairings never seen in training scored $+0.1002$ [$+0.0661$, $+0.1368$] and held-out *drugs* came out null. Two corrections have since removed that claim. The targets were rebuilt so the generic is fitted after the split rather than over the holdout, and the comparator was corrected: the scramble arm the gap was measured against is not a null but a different drug’s learned signature, chosen to point away from the truth. The held-out-drug arm is what exposed it, by returning a positive result on a split where a positive result is impossible.

Measured against the neutral comparator on the rebuilt targets, trained pairings score $+0.1457$ and held-out pairings $+0.0449$ [-0.0074 , $+0.0972$] — so there *is* a memorisation premium ($+0.1008$, cluster-robust $p = 0.015$), and cross-context transfer is not established. It is not shown to be zero either: the interval reaches $+0.097$, and at $n = 250$ that is a limit of power. What does survive, on two tests that use no comparator at all and so cannot be reached by this defect, is that the model emits the signature of the drug it is named ($+0.374$ on held-out pairings against $+0.390$ on trained ones) while scoring at chance when ranked against other drugs in the same cell line. The map exists and is coarse: the model learned the drug main effect and not the drug $\times$ cell-line interaction — which is also why the lookup below beats it, since the lookup *is* that main effect, measured directly.

But a lookup table wins — and it is not the ceiling either. A per-drug mean residual measured in other cell lines scores 0.963 against a 0.968 replicate ceiling; the model reaches 0.639. The natural reading of those three numbers is that the drug’s response is essentially a per-drug constant, that a conditional model therefore has almost nothing to condition on, and that the correct formulation of the task is retrieval rather than generation. We initially drew exactly that conclusion, and it is wrong.

It is wrong because NIR compresses near the top of its range — the same class of failure this thesis opens by documenting, arriving a second time in our own preferred instrument, at the ceiling instead of the floor. Measured in cosine space with measurement noise divided out, the transfer coefficient is $\mathcal{T}=0.557 [0.513, 0.601]$ against a simulated no-interaction null of 1.001, with a structure-matched negative control at +0.000. Roughly 45% of the drug-specific signal is a drug $\times$ cell-line interaction. The per-drug constant is what the lookup captures and what the model gropes toward; the interaction is reached by neither, and by nothing else in the literature we are aware of.

So the honest statement of what the model does is sharper, and less flattering, than “it misses the interaction”. It fails to recover the *main effect* that a numpy dictionary obtains for free, and the interaction is a further prize that no method here — ours least of all — comes near. Two different failures, and collapsing them undersells both.

That conclusion connects this work to an assumption the field has carried since the Connectivity Map: that a compound induces a characteristic signature stable enough across cellular contexts for signature-similarity querying to be meaningful. This thesis measures that assumption directly, with a calibrated metric and a proper replicate ceiling, at single-cell resolution — and the modern finding that a per-perturbation mean is hard to beat may be less a warning about weak baselines than a restatement of it.

It also places this work as the single-cell, generative instance of DrEval’s critique [13]. DrEval audits drug *sensitivity* prediction and finds that deep models barely outperform a naive additive predictor of mean drug and cell-line effects. Our `drug_lookup` is that predictor’s drug term, and the finding replicates on a 946-dimensional transcriptomic response: 0.963 against a 0.968 replicate ceiling, versus 0.639 for the model. Three of DrEval’s six pathologies — pseudoreplication, biased evaluation metrics, and missing ablation studies — turn out to structure the entire methodology of this thesis, which arrived at clustered bootstraps over cell lines, a discrimination metric immune to between-drug variance, and a token-level scramble ablation independently. That convergence is worth recording: the controls a critical benchmark demands are the same controls a single negative result needs in order to mean anything.

Where the opportunity actually is. Two further measurements locate it precisely, and between them they redirect what follows from this work.

The drug $\times$ cell-line interaction is large — roughly 45% of the drug-specific variance —

and we could not find structure in it. No cell line modulates different drugs in a consistent direction (excess within-line consistency -0.007 , and at plate scope an interval excluding anything above $+0.002$), and a direct test of whether mechanism-matched drugs share an interaction with a given cell line returned nothing detectable (permutation $p \approx 0.15\text{--}0.31$).

Those are two negative results and they do not license the conclusion that the interaction is irreducible. The second test bounds its own power severely: the interaction’s split-half reliability in this atlas is 0.195 , so κ is barely estimable, and same-mechanism compounds are so heavily co-plated that only 3–7% of matched pairs survive a different-plate requirement. The honest statement is that the structure of the interaction is *unresolved here*, and that resolving it needs more cells per condition or a plating design that crosses mechanism with plate — neither of which is an analysis choice. What can be said without qualification is that a lookup scoring 0.963 against a 0.968 ceiling is already close to what a per-drug constant can achieve, so on seen drugs nothing we built beats retrieval.

But a lookup requires the drug to have been seen. For drugs that have not been, protein-target and mechanism annotations score above a *count-matched* null ($+0.084$ [$+0.059, +0.110$] and $+0.078$ [$+0.056, +0.103$]), while chemical structure does not. That null is matched on the number of partners but not on plate: drugs sharing a mechanism are largely laid out on the same plate, drugs drawn at random are not, and the residuals being averaged retain plate structure. The arm and its null therefore differ in two respects where the estimand allows one, so an unknown share of these gaps is plate rather than annotation. A plate-matched re-measurement is pending and these two numbers should not be read as established until it lands. This overturns a pre-registration made earlier in this project, which inferred that unseen drugs were unreachable from a gate that tested chemical *structure* — the channel our own drug-knowledge specification ranked lowest — and never read the target annotations sitting in the atlas’s own metadata.

So the thesis does not end in closure. It ends with the opportunity located: on seen drugs a generative model has nothing to add over retrieval, because the residual part is unpredictable; on unseen drugs, where retrieval is impossible by construction, drug-side knowledge works and a model can do something a dictionary cannot. That is also the setting the benchmark literature calls Leave-Drugs-Out, and the one that matters for drug design.

A note on method. The substantive results here rest on controls that were added in response to specific objections: plate-matched comparison sets after a zero-information baseline more than doubled its score when the plate was free to vary; a swap-distance sweep after it was pointed out that scrambling to a similar drug is a weak test; a removal gate before trusting any ablation null; matched-norm injection before reading a swap; clustered bootstraps over cell lines rather than conditions; and a pre-registered interpre-

tation written before the decisive result, whose middle branch is the one that fired. Several claims were retracted along the way and are struck rather than deleted. In a literature where the central problem is that uninformative baselines score well, the controls are not an appendix to the result — they are most of it.

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A Appendix

A.1 Estimator validation against planted ground truth

Three of the measurements in this thesis rest on estimators built for the purpose rather than taken off the shelf. Each was validated by planting a known answer and checking recovery — in both directions, since an estimator that only ever returns the answer one hopes for is not evidence.

Estimator	Planted truth	Recovered	
transfer coefficient $\mathcal{T}$	$\mathcal{T} = 1.00$ (no interaction)	0.999	
	$\mathcal{T} = 0.70$	0.703	
	$\mathcal{T} = 0.40$	0.404	
channel gate	a channel that genuinely predicts	+0.397 (sd 0.029)	over 24 worlds
	a channel carrying nothing	−0.016 (sd 0.091)	over 24 worlds
κ -structure	a consistent per-line modulation	excess +1.085	
	an idiosyncratic interaction	excess +0.001	

Table 16: All three recover planted answers in both directions.

Two of these validations changed the estimators rather than merely endorsing them. The channel gate’s count-matched null originally drew its random partners from a pool *excluding* the channel’s own selections, which depletes the null of exactly the partners the channel chose and anti-correlates the two arms. And the useless-channel gap has a standard deviation of roughly 0.09 on a single small world, so the original validation — asserting that one draw fell within 0.10 — tested luck rather than correctness; it now averages 24 independent worlds. That variance is also why the reported channel results are gated on a clustered interval rather than on a point estimate.

A.2 Absent-gene convention

A predicted cell sentence lists only expressed genes, so some panel genes are unranked and a convention is required. Two were carried throughout: **worst** assigns absent genes the last rank P , **francesca** a fixed middle rank $P/2$. The choice does not matter — spike-in DE- Δr reads 0.952 under the first and 0.954 under the second, and no ceiling or baseline in this thesis moves by more than 0.01 between them. All figures in the body are quoted under **worst**.

A.3 Swap-distance sweep

The stratified scramble of Section 3.4 was preceded by a continuous version: Δ_{scr} measured as a function of the response distance between the real and the swapped-in drug, binned

into quintiles.

Bin (swap distance)	single-cell model	optimal-transport model
0 — nearest (2.75–4.03)	−0.046	−0.001
1 (4.03–4.60)	+0.060	+0.031
2 (4.60–4.93)	−0.006	+0.017
3 (4.93–5.32)	−0.004	+0.043
4 — farthest (5.32–6.25)	−0.019	−0.005
trend, corr(distance, Δ_{scr})	+0.051	+0.012

Table 17: No bin in either model has an interval excluding zero, and the trend correlations are ≈ 0 . 200 (drug, swap) pairs over 20 cell lines, $n = 40$ per bin.

Two features make this an argument rather than five nulls. The single borderline value sits in a *middle* bin, whereas a genuine dissimilarity effect would be monotone and peak in the far bin. And the comparison is paired: the model’s own NIR drifts across bins from 0.497 to 0.326, because drugs that are far from everything populate the far bins, but Δ_{scr} uses the same drug on both arms so that drift cancels.

The intervals are wide — $n = 40$ per bin gives half-widths of 0.06–0.10 — so this sweep excludes a large effect rather than establishing a null, which is why the stratified scramble replaced it in the body.

A.4 Training configuration

Every arm was cold-started from the same base checkpoint with identical hyperparameters, so that the target or objective is the only variable. One exception is material and is stated wherever the arm is quoted.

Arm	Target	Training	Note
single cell	the observed treated cell	1 epoch	
consensus	group pseudobulk	$\approx 60\%$ of an epoch	tier-1 NIR unscored
optimal transport (T2)	barycentric, $\varepsilon = 0.5$	1 epoch, 23,842 steps	
residual	signed residual signature	1 epoch, 15,340 steps	
residual (holdout)	as above, 738 conditions withheld	1 epoch	

Table 18: Learning rate 10^{-5} , batch size 1, gradient accumulation 16, maximum length 8192, bf16, weight decay 0.01, warmup ratio 0.03 throughout. The consensus arm stopped early on a flat loss; since it is one endpoint of the ε -ladder, the ladder’s negative rests on the other two.

A.5 Calibration self-test

The calibration framework of Section 3.3 is itself checked, since a metric audit whose instrument is untested proves nothing. Beyond deterministic unit checks — a perfect prediction must reach the metric’s maximum, a zero shift and a flat truth must return no score — the self-test asserts the exploit under a *second, independent* zero-information baseline: a leave-one-out mean scores $\text{DE-}\Delta r = 1.000$ while scoring $\text{NIR} = 0.000$. Two unrelated constructions both saturating the standard metric, and both scoring zero on the discrimination metric, establishes the exploit as a property of the metric rather than of any one baseline.